CONFOEDERATIO,
RESEARCH DIVISION (CRD).

MUSCOVY, THE POLISH-LITHUANIAN FRONTIER AND THE CONQUEST OF SIBERIA

Population and Towns across Russia, Ukraine, Belarus, Lithuania and the Baltic

11 November 2026 - Aust Kätzchen | austkaeztchen@confoederatio.org

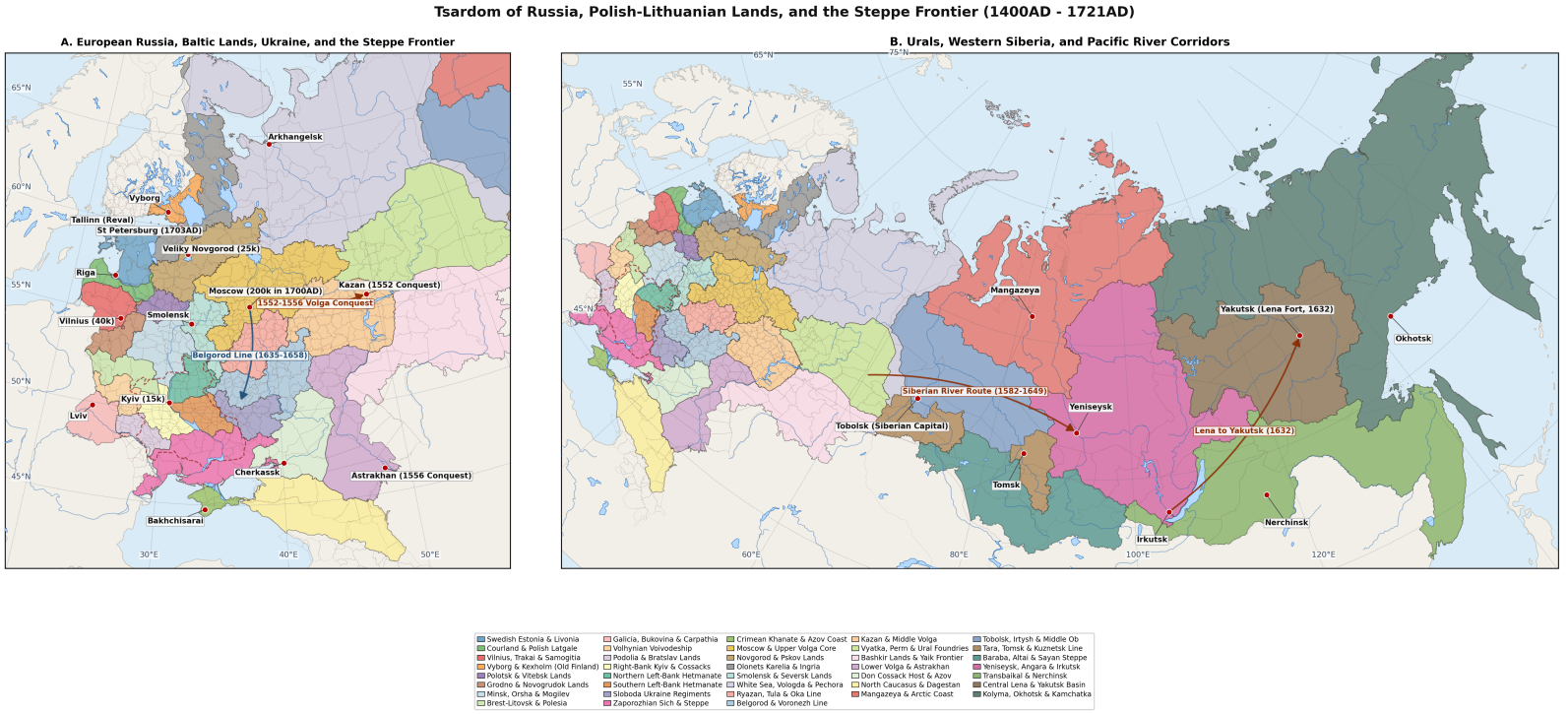


Figure 1. The 38 historical regions and principal towns across Eastern Europe and Northern Asia from 1400AD to 1721AD, showing the western Baltic and Polish-Lithuanian palatinates, the Muscovite forest core, the southern abatis defence lines and the eight Siberian river-portage zones.

Abstract

Between 1400AD and the Treaty of Nystad in 1721AD, the lands between the Baltic Sea and the Pacific Ocean were remade by the rise of Muscovy, the union and wars of Poland-Lithuania and the Russian crossing of the Urals. In 1400AD the total population across the 38 historical regions of the Baltic provinces, Lithuania, Belarus, Ukraine, Crimea, Russia, the North Caucasus and Siberia stood at 11,36M: 935k in the Baltic provinces, Lithuania and Vyborg-Kexholm (Regions 1-4), 950k in the Belarusian palatinates (Regions 5-8), 2,51M across Ukraine and Crimea (Regions 9-17) and 6,96M in Russia and Siberia (6,66M in European Russia, the Steppe and the North Caucasus across Regions 18-30, and 300k in Siberia across Regions 31-38) [1][2][3][4][5][6]. Over the next three centuries the population nearly doubled to 20,7M on the eve of the mid-seventeenth-century wars in 1648AD and reached 22,83M at Peter the Great's First Soul Revision in 1720AD: 1,26M in the Baltic, Lithuania and Old Finland, 1,55M in Belarus, 5,65M in Ukraine and Crimea, and 14,37M across Russia and Siberia (13,61M in European Russia

and 760k in Siberia) [7][8][9][10][11].

Two opposite movements produced this result. On one side, war, famine and plague repeatedly battered the western borderlands and the old forest heartland: the Livonian War and Ivan IV's *Oprichnina* (1558-1583AD), the famine of 1601-1603AD and the Time of Troubles (*Smuta*, 1604-1618AD), the Russo-Polish Deluge (*Potop*, 1654-1667AD) across Lithuania and Belarus, *The Ruin* (*Ruina*, 1657-1687AD) in Right-Bank Ukraine, the Great Famine of 1695-1697AD in Estland, Livland and Finland, and the Great Northern War and plague of 1708-1711AD [12][13][14][15][16]. On the other side, steady peasant and Cossack settlement filled the Left-Bank Hetmanate, Sloboda Ukraine, the Oka, Belgorod and Simbirsk defence lines (*Zasechnye Cherty*), the Middle Volga, the Urals and the river routes and southern ploughlands of Siberia [17][18][19][20].

Background

Between the fifteenth and early eighteenth centuries, the 18,12M km^2 expanse of Eastern Europe and Northern Asia (excluding Central Asia and the South Caucasus, which entered the Russian Empire only in the eighteenth and nineteenth centuries) belonged to several rival states whose borders governed how people settled and how taxes were counted. To follow these shifting frontiers without losing continuity, we divide the territory into 38 historical regions grouped into five main geographical divisions (Figure 1 and Table 1).

Along the western littoral, the four regions of the Baltic, Lithuania and Old Finland (Regions 1-4, 229,08k km^2) covered the German knightly and bishopric lands of *Terra Mariana* (later Swedish Estland and Livland), the Duchy of Courland and Polish Inflanty, the Lithuanian heartland of Vilnius, Trakai and Samogitia, and the Karelian border fiefs of Vyborg and Kexholm. Favoured by Baltic grain and flax exports, this western rim nearly doubled from 935k in 1400AD to 1,79M in 1648AD, before the mid-century Deluge, the famine of 1695-1697AD and the Great Plague of 1710-1711AD cut the population back to 1,26M in 1720AD. Immediately to the south-east, the four Belarusian regions of the Grand Duchy of Lithuania (Regions 5-8, 246,65k km^2) - Polotsk and Vitebsk on the Western Dvina, Grodno and Novogrudok in Black Ruthenia, Minsk and Mogilev along the Upper Dnieper, and Brest-Litovsk and Pinsk in Polesia - more than doubled from 950k in 1400AD to 2,5M in 1648AD, only to lose nearly half their people in the war of 1654-1667AD and stand at 1,55M in 1720AD.

Further south, the nine Ukrainian and Crimean regions (Regions 9-17, 607,83k km^2) stretched from the Polish Crown palatinates of Galicia, Volhynia, Podolia and Right-Bank Kyiv across the Dnieper to the Left-Bank Hetmanate, Sloboda Ukraine, the Zaporozhian Wild Fields and the Crimean Khanate. Together their population more than doubled from 2,51M in 1400AD to 6,1M in 1648AD. In the decades of *The Ruin* after 1657AD, tens of thousands of families fled the devastated Right Bank (where Kyiv, Podolia and Volhynia dropped from 2,75M in 1648AD to 1,53M in 1720AD) to settle east of the Dnieper in the Left-Bank Hetmanate and Sloboda Ukraine, which tripled from 1,11M in 1648AD to 1,95M by 1720AD.

Within the Tsardom of Russia itself, the thirteen European, Volga-Ural, Steppe and North Caucasian regions (Regions 18-30, 4,24M km^2) began in 1400AD with 6,66M people concentrated in the northern forest belt of Moscow, Novgorod-Pskov, Smolensk and Pomorye (3,76M across Regions 18-22). After the conquest of Kazan (1552AD) and Astrakhan (1556AD) and the building of the Oka, Belgorod and Simbirsk fortified lines, peasant settlement poured southward into the black-earth forest-steppe and eastward to the Volga and the Urals, carrying the total across Regions 18-30 to 9,87M in 1648AD and 13,61M in 1720AD. Beyond the Urals, the eight vast zones of Siberia (Regions 31-38, 12,81M km^2) held roughly 300k native hunters, fishers and herders in 1400AD; after Yermak's crossing in 1582AD, Russian fur-trappers, garrison men and plough peasants pushed along the river portages to the Pacific, lifting Siberia's population from 338k in 1580AD to 435k in 1648AD and 760k (nearly half of them Russians) by 1720AD.

Historical Region/ Subtotal (AD)	1400	1500	1550	1580	1600	1615	1648	1667	1678	1700	1720
1. Terra Mariana/ Swedish Estland & Livland	260k	330k	390k	270k	290k	260k	360k	340k	390k	440k	300k
2. Courland, Semigallia & Polish Inflanty	140k	175k	210k	190k	210k	200k	260k	210k	240k	280k	200k
3. GDL Core: Vilnius, Trakai & Samogitia	460k	610k	720k	780k	850k	890k	1,06M	640k	710k	810k	650k
4. Vyborg Fief & Kexholm/ Old Finland	75k	92k	105k	88k	95k	80k	110k	105k	118k	130k	110k
5. Polotsk & Vitebsk Voivode-ships (GDL)	240k	320k	390k	350k	410k	440k	560k	270k	310k	360k	370k
6. Grodno & Novo-grudok Voivode-ships (GDL)	280k	385k	470k	520k	580k	620k	740k	420k	470k	520k	490k
7. Minsk, Orsha & Mogilev Dnieper (GDL)	260k	350k	430k	460k	510k	550k	680k	330k	380k	430k	430k
8. Brest-Litovsk & Pinsk-Turov Polesia (GDL)	170k	245k	310k	340k	380k	410k	520k	260k	290k	320k	260k
9. Ruthenian/ Galicia, Belz & Carpathian Rus	820k	1,02M	1,18M	1,29M	1,42M	1,51M	1,75M	1,38M	1,45M	1,58M	1,62M

10. Volhynian Voivode-ship (Lutsk & Ostroh)	420k	530k	620k	680k	750k	790k	920k	580k	610k	660k	680k
11. Podolian & Bratslav Voivode-ships	350k	450k	540k	610k	690k	740k	910k	490k	380k	440k	510k
12. Right-Bank Kyiv Regiments (Chyhyryn)	280k	360k	440k	520k	620k	690k	920k	520k	360k	320k	340k
13. N. Left-Bank Hetmanate (Chernihiv-Nizhyn)	210k	260k	300k	330k	370k	390k	490k	560k	620k	720k	790k
14. S. Left-Bank Hetmanate (Poltava-Lubny)	110k	145k	180k	220k	270k	320k	460k	540k	610k	710k	760k
15. Sloboda Ukraine Regiments (Kharkiv-Sumy)	50k	65k	80k	95k	110k	120k	160k	250k	310k	380k	400k
16. Zaporozhian Sich & Pontic Wild Fields	60k	70k	80k	90k	100k	105k	130k	135k	145k	165k	180k
17. Crimean Khanate & Ottoman Kefe	210k	260k	290k	310k	330k	335k	360k	365k	375k	385k	370k
18. Moscow & Zamoskovny Core (Vladimir-Tver)	1,95M	2,62M	3,05M	2,78M	2,95M	2,28M	2,85M	3,08M	3,32M	3,62M	3,72M
19. Novgorod & Pskov Five Pyatiny	920k	1,17M	1,34M	960k	1,04M	710k	930k	1,01M	1,09M	1,18M	1,23M

20. Olonets Karelia & Baltic Ingria (St P.)	200k	250k	280k	220k	240k	180k	220k	230k	250k	270k	350k
21. Smolensk & Seversk Border-lands (Bryansk)	480k	600k	680k	640k	690k	510k	640k	580k	640k	710k	730k
22. Pomorye, Vologda, Dvina & Pechora North	210k	280k	340k	380k	410k	420k	460k	470k	480k	450k	420k
23. Ryazan, Tula, Oryol & Great Oka Abatis	680k	850k	980k	940k	1,02M	820k	1,08M	1,18M	1,29M	1,42M	1,52M
24. Belgorod & Sevsk Razryads (Voronezh-Kursk)	190k	240k	290k	340k	410k	380k	620k	820k	980k	1,22M	1,44M
25. Kazan, Nizhny Novgorod & Simbirsk Line	780k	890k	940k	960k	1,02M	980k	1,18M	1,32M	1,46M	1,62M	1,76M
26. Vyatka, Great Perm & Stroganov Urals	220k	280k	330k	360k	390k	405k	480k	520k	560k	610k	680k
27. Bashkir Four Darugas (Ufa) & Yaik Frontier	180k	200k	215k	225k	235k	240k	260k	275k	290k	310k	340k
28. Lower Volga, Astrakhan & Kalmyk Steppe	240k	260k	270k	265k	275k	280k	320k	350k	375k	400k	430k
29. Don Cossack Host (Cherkassk) & Azov	90k	100k	110k	120k	135k	130k	165k	190k	215k	240k	250k

30. North Caucasus (Circassia, Kabarda, Dagestan)	520k	565k	595k	610k	625k	635k	665k	680k	695k	715k	740k
31. Pomor-Mangazeya Maritime & Taz Portage	16k	17k	18k	19k	24k	28k	26k	22k	19k	18k	18k
32. Tobolsk-Irtysh-Middle Ob Portage Highway	42k	45k	47k	49k	56k	64k	88k	105k	118k	138k	158k
33. S. Tobol-Tara-Tomsk-Kuznetsk Ploughland	38k	42k	45k	47k	52k	60k	86k	118k	145k	195k	254k
34. Nomadic Baraba-Altai-Minusinsk Steppe	74k	77k	80k	81k	82k	82k	84k	82k	80k	88k	100k
35. Ket-Yeniseysk-Angara-Ilimsk-Irkutsk Spine	28k	30k	31k	32k	33k	34k	44k	56k	66k	82k	98k
36. Transbaikal Dauria & Upper Amur (Nerchinsk)	34k	36k	37k	38k	39k	39k	40k	44k	48k	55k	62k
37. Central Lena-Vilyuy-Aldan (Yakutsk Alaas)	38k	40k	41k	42k	42k	41k	42k	43k	44k	45k	47k
38. NE Arctic & Pacific Yasak (Kolyma-Kamchatka)	30k	30k	31k	30k	29k	27k	25k	25k	25k	24k	23k

Baltic, Lithuania & Old Finland (Reg. 1-4)	935k	1,21M	1,43M	1,33M	1,45M	1,43M	1,79M	1,29M	1,46M	1,66M	1,26M
Belarus GDL Subtotal (Reg. 5-8)	950k	1,3M	1,6M	1,67M	1,88M	2,02M	2,5M	1,28M	1,45M	1,63M	1,55M
Ukraine & Crimea Subtotal (Reg. 9-17)	2,51M	3,16M	3,71M	4,14M	4,66M	5M	6,1M	4,82M	4,86M	5,36M	5,65M
Russia European & Steppe (Reg. 18-30)	6,66M	8,3M	9,42M	8,8M	9,44M	7,97M	9,87M	10,7M	11,64M	12,77M	13,61M
Russia Siberian Zones (Reg. 31-38)	300k	317k	330k	338k	357k	375k	435k	495k	545k	645k	760k
Russia Subtotal (Reg. 18-38)	6,96M	8,62M	9,75M	9,14M	9,8M	8,35M	10,3M	11,2M	12,19M	13,41M	14,37M
Total Domain (All 38 Regions)	11,36M	14,29M	16,48M	16,28M	17,78M	16,8M	20,7M	18,59M	19,96M	22,06M	22,83M

Table 1. Regional population estimates across all 38 regions and five main geographical groupings from 1400AD to 1720AD, showing the contrast between mid-seventeenth-century wartime losses in the west and steady frontier growth in the south and east.

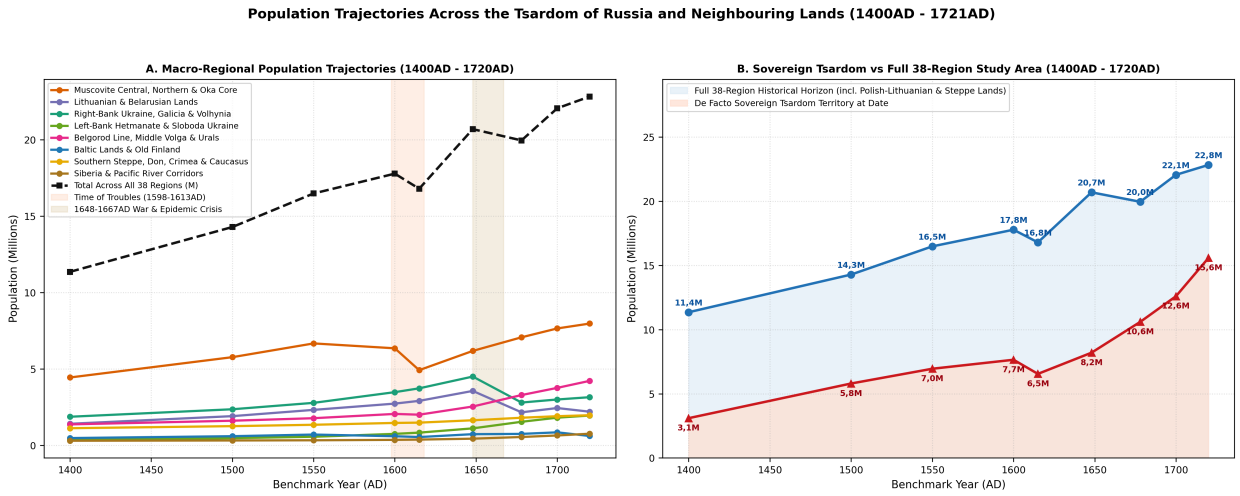


Figure 2. Population totals across the five main geographical divisions from 1400AD to 1720AD, contrasting the steady expansion of the Russian forest-steppe, Volga-Ural and Siberian frontiers with the deep wartime drops in Belarus, Lithuania, Right-Bank Ukraine and the Baltic during the Deluge and the Great Northern War.

Methodology

Population estimates across the 38 regions between 1400AD and 1721AD rest on six main tax and church registration systems that gradually moved from counting ploughland units to counting households, hearths and finally individual male souls [3][7][21][22]. In the Muscovite heartland and northern provinces (Regions 18-26), late fifteenth- and sixteenth-century scribes compiled land books (*pistsovye knigi*) that assessed dues by ploughland units - the Muscovite *sokha* (400 to 800 *chetverti* of good arable land across three fields, adjusted for service, church or state-peasant tenure) and the Novgorodian *obzha* (Regions 19-20) [2][23]. Because peasants responded to wartime taxes by leaving assessed ploughland or crowding several nuclear families under a single roof, household (*dvor*) counts from the land books must be multiplied by observed household sizes of 6 to 8,5 people [1][19].

When the Tsardom of Russia switched from land taxation (*pozemelnoe*) to household taxation (*podvornoe*), it carried out the general household counts of 1646AD and 1678AD (*perepisnye knigi*) [7][21]. By comparing the 1678AD household books against the 1710AD wartime audit and Peter I's First Soul Revision (1719-1721AD, checked by military audit commissions in 1721-1724AD), Ya. E. Vodarsky (1977) and V. M. Kabuzan (1963) showed that the seventeenth-century household books left out rural priests, military servicemen (*sluzhilye lyudi*) and landless *bobyli* or *zadvornye* cottagers, while encouraging families to merge into large compounds of 8,5-9,2 people (4,2-4,6 males) per *dvor* [7][8]. Replacing the household tax with a head tax on every male (*podushnaya podat*) in 1718-1719AD removed the reason to hide split families and counted 7,18M male souls (14,37M people of both sexes across Regions 18-38 in 1720AD) [8][24].

Along the Baltic coast and in Karelia (Regions 1, 2 and 4), population figures are based on Livonian and Swedish ploughland audits (*Hakenrevisionen* of 1586-1591AD, 1601AD, 1638AD, 1688AD and 1712AD), Lutheran parish books (*kirikuraamatud* kept across Estland, Livland and Vyborg-Kexholm from the mid-seventeenth century), Courlandian ducal estate inventories and Swedish poll-tax rolls (*mantalslangder*) studied by H. Palli (1996), S. Vahtre (1973), E. Dunsdorfs (1962) and V. Saloheimo (1976) [3][10][11][25]. In Lithuania (Region 3), Belarus (Regions 5-8) and Polish Crown Ukraine (Regions 9-12), estimates draw on the 1528AD Grand Duchy of Lithuania muster roll (*Popis voyska*, listing 19,8k noble

horsemen at a legal rate of 1 horseman per 8 peasant *sluzhby* alongside grand-ducal and church estates), the 1557AD *Voloka* land survey (*Pomiara wloczna*), Crown estate inspections (*lustracje* of 1564-1565AD, 1616AD and 1629AD) and hearth-tax rolls (*podymne*) from 1629AD, 1648-1650AD, 1667AD, 1690AD and 1717AD [4][5][9][13][26]. Finally, figures for the Cossack Hetmanate and Sloboda Ukraine (Regions 12-15), Crimea (Region 17) and Siberia (Regions 31-38) come from the 1649AD Zboriv Register, the 1666AD Left-Bank census books (*perepysni knyhy*), Ottoman tax registers (*tahrir defterleri*) for Kefe and Siberian fur-tribute (*yasachnye*) and inspection (*dozornye*) books [6][20][27][28].

Source Group	Period	Tax or Census Unit	Household Multiplier	Regional Coverage	Key Studies
Baltic Hakenrevisionen & Mantalslangder	1561-1715AD	Adramaa/ Haken, farmstead (talu) & parish soul	7,5-9,5 persons/talu farmstead	Regions 1, 2 & 4 (Estland, Livland, Courland, Vyborg)	Palli 1996; Vahtre 1973; Dunsdorfs 1962; Saloheimo 1976
Lithuanian 1528AD Popis & Voloka Books	1528-1600AD	Cavalry horse, sluzhba & voloka (21,36 ha)	6,5-7,5 persons/ voloka-sluzhba	Regions 3, 5-8 & 10 (Lithuania, Belarus & Volhynia)	Lowmianski 1983; Ochmanski 1982; Vasiliauskas 2012
Polish Crown Lustracje & Podymne Hearths	1564-1717AD	Taxable hearth (dym) & lan arable unit	6-6,5 persons/ rural dym	Regions 2-3 & 5-12 (Inflanty, GDL & Crown Rus)	Morzy 1965; Karpachev & Kozlovsky 1972; Yakovenko 2006
Novgorod & Moscow Pistsovye Knigi	1495-1594AD	Sokha, obzha & peasant household (dvor)	5,5-7 persons/ dvor	Regions 18-23 (Zamoskovny, Pyatiny, Ingria, Oka)	Kopanev 1959; Shapiro 1971, 1974; Selin 2020
Zboriv Register, 1666AD Books & Tahrir	1520-1700AD	Cossack kurin/ dvor, posad dvor & Ottoman hane	5,5-7 persons/ household	Regions 12-17 (Hetmanate, Sloboda & Crimea)	Krypiakevych 1954; Irtenina 2022; Ozturk 2019
1646AD & 1678AD Perepisnye Knigi	1646AD; 1678AD	Taxable household (tyaglyy dvor)	8-9,2 persons/ enlarged dvor	Regions 18-29 (Muscovite Core & Frontier Razryads)	Vodarsky 1973, 1977; Avdeev 2025
Siberian Yasachnye & Dozornye Knigi	1595-1719AD	Yasak-paying adult male (18-50 years)	4-4,2 total pop/ yasak payer	Regions 31-38 (8 Siberian Portage & Yasak Zones)	Dolgikh 1960; Pokrovsky 1984; Ushnitskiy 2017b
First Poll-Tax Revision (Revizskie Skazki)	1719-1724AD	Individual male soul (revizskaya dusha)	2-2,02 total pop/male soul	Regions 1, 4, 13-15 & 18-38 (Petrine Empire)	Kabuzan 1963, 1990; Ulyanova 2016; Mironov 2007

Table 2. Principal tax rolls, land surveys, parish registers and household censuses from 1400AD to 1721AD, with their assessment units and household multipliers.

Results

The Baltic, Lithuania and Old Finland

Estland, Livland and Courland

Across the Eastern Baltic littoral (Regions 1 and 2, covering 113,51k km^2 combined), late medieval political authority belonged to the ecclesiastical and knightly confederation of *Terra Mariana* (the Archbishopric of Riga, the Bishoprics of Dorpat, Osel-Wiek and Courland, and the Livonian Branch of the Teutonic Order) alongside Hanseatic mercantile municipalities led by Riga (8k in 1400AD to 12k in 1550AD), Reval/Tallinn (5,5k to 8k), Dorpat/Tartu (4,5k to 6,5k), Pernau/Parnu (1,6k to 2,6k), Narva (1,8k to 3,2k), Wenden/Cesis (1,5k to 2,4k) and Arensburg/Kuressaare [3][10][29]. Favoured by a century of relative internal peace after 1435AD and rising rye, flax and timber exports to Western Europe, Terra Mariana/Estland and Livland (Region 1, 76,23k km^2) grew from 260k in 1400AD to 330k in 1500AD and 390k in 1550AD, while southern Courland, Semigallia and Latgale (Region 2, 37,28k km^2) rose from 140k to 210k [3][10][25].

Ivan IV's invasion of Narva and Dorpat in 1558AD shattered the medieval confederation and initiated seventy years of intermittent Baltic warfare. During the Livonian War (1558-1583AD) and the subsequent Polish-Swedish wars of 1600-1629AD - compounded by the extreme volcanic-winter crop failure, famine and plague of 1601-1603AD - rural *adramaa* farmstead occupancy across mainland Estland and northern Livland dropped by more than half [3][14]. Region 1 fell by 30,8% between 1550AD (390k) and 1580AD (270k) and stood at 260k in 1615AD before the Truce of Altmark (1629AD) and Treaty of Bromsebro (1645AD, ceding Osel/Saaremaa from Denmark) consolidated Swedish Estland and Livland [3][30]. Under 17th-century Swedish administration, inward rural migration of Finnish, Russian, Latvian and Lithuanian settlers, parish schooling and Charles XI's *Reduktion* of noble estates in the 1680sAD drove demographic recovery. Region 1 climbed to 360k in 1648AD, 390k in 1678AD and 440k at the turn of the century (c. 1695-1700AD), despite the severe Great Famine of 1695-1697AD (*Suur nalg*), when cold, rain-soaked summers destroyed two consecutive harvests and killed roughly 70k-75k inhabitants across Estland and Livland while Swedish authorities continued shipping Baltic manor grain to Stockholm and Finland [3][14][31].

South of the Daugava (Western Dvina), the Pact of Vilnius (1561AD) secularised the southern Ordensstaat into the hereditary Duchy of Courland and Semigallia under Gotthard Kettler as a fief of the Polish-Lithuanian Commonwealth, while eastern Latgale remained under direct Commonwealth sovereignty after 1629AD as the Voivodeship of Inflanty (Dunaburg/Daugavpils, Rezhitsa/Rezekne and Ludza) [10][32]. Shielded from the worst fighting of the Livonian War (dipping mildly from 210k in 1550AD to 190k in 1580AD), Region 2 expanded to 260k in 1648AD under the mercantilist manufactures, shipyards and Atlantic-Baltic trade of Duke Jacob Kettler (1642-1682AD) at his capital Mitau/Jelgava (5,5k in 1648AD), the shipbuilding port of Windau/Ventspils (3,2k), Libau/Liepaja (3,5k), Goldingen/Kuldiga (3k) and Bauske (2,5k) [10][33]. After suffering a 19,2% contraction (260k to 210k) during the 1658-1660AD Swedish occupation of Courland and Muscovite capture of Dunaburg (briefly renamed *Borisoglebsk* in 1656-1666AD), Region 2 recovered to 280k by 1700AD [10][34].

Both Baltic regions then suffered heavy losses during the Great Northern War (1700-1710AD) and the Great Baltic Plague of 1710-1711AD [3][15][33][34]. In Estland and Livland (Region 1), the population plunged by 43,2% in a single decade, from 440k in 1700AD to 250k in 1710AD, as Russian and Swedish campaigns around Narva (1700AD and 1704AD), Dorpat (besieged in 1704AD and demolished in 1708AD with its burghers deported to Vologda), Pernau and Riga culminated in the *Yersinia pestis* epidemic of summer-autumn 1710AD. That outbreak killed over half the besieged population of Riga (15k to 6,5k), roughly two-thirds of Reval (5k to 4,5k) and up to 70% of western Estland coastal parishes before the Baltic knighthoods signed the Capitulations of 1710AD, confirmed by the Treaty of Nystad in 1721AD as the

regional total recovered to 300k in 1720AD [3][15][30]. Across Courland, Semigallia and Polish Inflanty (Region 2), the population fell by 39,3%, from 280k in 1700AD to 170k in 1710AD, before recovering to 200k by 1720AD; Saxon, Swedish and Russian armies fought across the Daugava (1701AD) and Gemauerthof (1705AD), and the 1710AD plague wiped out nearly half of Courland's rural inhabitants while reducing Mitau from 6,8k to 2,6k [10][33][34].

Core Lithuania

Immediately south of Courland and Livonia lay the political and ethnic Lithuanian core of the Grand Duchy of Lithuania in Region 3 (65,88k km^2), uniting the western districts of the Vilnius and Trakai Voivodeships with the Eldership of Samogitia (*Zemaitija*) [4][35]. Untouched by Tatar steppe raiding north of the Neman (Nemunas) and stimulated by German/Prussian and Baltic river commerce through the Hanseatic factory at Kaunas, Region 3 grew steadily from 460k in 1400AD to 610k in 1500AD and 720k in 1550AD. In Henryk Lowmianski's and Jerzy Ochmanski's reconstructions of the 1528AD GDL Military Census (*Popis voyska*), ethnic Lithuania and Samogitia exhibited the highest rural settlement densities in the entire Grand Duchy [4][26]. Following Sigismund II Augustus's 1557AD Volok Reform (*Valaku reforma*), which reorganised grand-ducal and magnate domains (such as the Siauliai Crown Economie) into regular three-field *voloka* villages and chartered scores of market towns (*miesteliai*), Region 3 expanded to 850k in 1600AD and 1,06M in 1648AD [35][36]. At its mid-17th-century peak, the Grand-Ducal capital Vilnius (seat of the 1579AD Jesuit University, the Lithuanian Tribunal and thriving Ruthenian, Polish, Lithuanian, Jewish, German and Karaite communities) reached ~28k inhabitants, flanked by Kaunas (~15k), the Radziwill Protestant commercial hub of Kedainiai (~6,5k), Trakai (~5k), Siauliai (~3,5k), Raseiniai (~3,2k, parliamentary seat of Samogitia) and Telsiai (~2,6k) [35][36][37].

This growth ended in August 1655AD when Tsar Aleksey Mikhailovich's armies and Ivan Zolotarenko's Cossacks captured Vilnius - the first foreign capture of the Lithuanian capital in its history - sparking a seventeen-day conflagration, while Swedish forces occupied Samogitia and Kedainiai [13][35]. Six years of Muscovite occupation in Vilnius and Trakai (1655-1661AD), famine, the 1657-1658AD plague and artisan deportations reduced Region 3 by 39,6%, from 1,06M in 1648AD to 640k in 1667AD (with Vilnius shrinking to 11,5k and Kaunas to 6k) [4][13]. Although post-war reconstruction lifted Region 3 back to 710k in 1678AD and 810k in 1700AD, the Great Northern War (during which Swedish, Russian, Saxon and rival Sapieha-Oginski Lithuanian confederate armies ravaged Samogitia and Vilnius) and the Great Plague of 1708-1711AD killed more than a quarter of the population, driving Region 3 down to 580k in 1710AD (-28,4%) before an initial recovery to 650k in 1720AD [16][32][35].

Vyborg and Kexholm

On the Karelian Isthmus, Lake Saimaa and the western shore of Lake Ladoga, Region 4 (49,69k km^2) bridges the Swedish-Finnish and Novgorodian-Russian worlds [11][38]. Throughout the fifteenth and early sixteenth centuries, the western half of this territory formed the Swedish border province of Vyborg Fief (*Viipurin linnalaani*), guarded by the stone castle of Vyborg/Viipuri (chartered 1403AD, 2,2k in 1400AD to 3,6k in 1550AD) and the inland Savo lake fortress of Nyslott/Savonlinna (*Olavinlinna*, founded 1475AD), while the eastern Ladoga shore belonged to Novgorod's and Muscovy's Korela (*Kexholm/Priozersk*) district [23][38]. Combined population grew from 75k in 1400AD to 105k in 1550AD before the 25-year Russo-Swedish "Long Wrath" (*Pitka viha*, 1570-1595AD) and the Swedish occupation of Kexholm (1580-1595AD) reduced regional numbers to 88k in 1580AD and 80k during the *Smuta* and the Ingrian War in 1615AD [38][39].

When the Treaty of Stolbovo (February 1617AD) handed Kexholm County (*Kakisalmen laani*) to Sweden, the population of Region 4 was almost completely replaced, a turnover documented in Swedish tax rolls and Muscovite land books by V. Saloheimo (1976), K. Katajala (2017), O. M. Fishman (2020) and Yu. V.

Stepanova (2024) [11][38][40][41]. Facing heavy Swedish crown rents, noble fief (*forlaning*) dues and Lutheran pressure under Bishop Isaac Rothovius's successors, Orthodox Karelian families trekked across the border into Russia - first during the 1620s-1640sAD (some 25k migrants) and then during the Russo-Swedish War of 1656-1658AD (*Ruptuuri*), when another 20k-25k Orthodox Karelians left with the Russian army and were settled by Tsar Aleksey Mikhailovich on crown lands in Bezhetskiy Verkh, Tver, Torzhok, Vyshny Volochyok and Valdai (Regions 18-19), founding the Tver Karelian communities [40][41]. In their place, Lutheran peasants from Savonia (*Savo*) and Vyborg Karelia moved into the empty Kexholm parishes to clear burn-beaten (*kaski*) rye plots, lifting Region 4 to 110k in 1648AD and 130k by 1700AD alongside new Saimaa and Gulf of Finland tar-shipping towns at Villmanstrand/Lappeenranta (1649AD) and Vehkalahti/Fredrikshamn (Hamina, 1653AD) [11][38]. The Finnish Great Famine of 1696-1697AD (*Suuret kuolonvuodet*), followed by Peter I's sieges of Vyborg and Kexholm in 1710AD and the Great Wrath occupation (*Isoviha*), cut Region 4 by 26,9% to 95k in 1710AD, before numbers recovered to 110k in 1720AD as Vyborg and Kexholm ("Old Finland", *Vanha Suomi*) passed to Russia at the Treaty of Nystad (1721AD) [11][38].

Western/Baltic Region	Area	Main Towns & Fortresses	1648AD/ 1700AD	Wartime Low (% Drop)	1720AD
1. Estland & Livland (Terra Mariana/ Ostsee)	76,23k km^2	Riga, Reval (Tallinn), Dorpat (Tartu), Narva, Pernau, Wenden, Arensburg	360k/440k	250k (1710AD, -43,2%)	300k
2. Courland, Semigallia & Polish Inflanty	37,28k km^2	Mitau (Jelgava), Libau, Windau, Goldingen, Bauske, Dunaburg, Rezhitsa	260k/280k	170k (1710AD, -39,3%)	200k
3. GDL Core: Vilnius, Trakai & Samogitia	65,88k km^2	Vilnius, Kaunas, Trakai, Kedainiai, Siauliai, Raseiniai, Telsiai	1,06M/ 810k	640k (1667AD)/ 580k (1710AD)	650k
4. Vyborg Fief & Kexholm/Old Finland	49,69k km^2	Vyborg (Viipuri), Kexholm (Priozersk), Nyslott, Fredrikshamn, Villmanstrand	110k/130k	80k (1615AD)/ 95k (1710AD)	110k
Western Rim Subtotal (Regions 1-4)	229,08k km^2	27 chartered Hanseatic, Magdeburg, ducal and Swedish staple towns	1,79M/ 1,66M	1,09M (1710AD, -34% vs 1700AD)	1,26M

Table 3. Land areas, main towns and population totals across the Baltic provinces, Lithuania and Old Finland from 1400AD to 1720AD, highlighting the losses of the 1654-1667AD Deluge and the 1710-1711AD plague.

The Muscovite Core and the Russian North

The Forest Heartland, Novgorod and Pomorye

At the opening of the 15th century, the five northern and western forest regions of Russia (Regions 18-22, totalling 3,76M in 1400AD) were recovering from recurrent waves of the Black Death (1352-1353AD, 1364-1366AD, 1417AD and 1425-1427AD) and the destructive campaigns of Tokhtamysh (1382AD) and Edigu (1408AD) [1][42]. Within the Moscow and Zamoskovny core (Region 18, 277,15k km^2, rising from 1,95M in 1400AD to 3,05M in 1550AD), internal agricultural colonisation advanced along the watershed

ridges (*vodorazdely*) of the Moscow-Klyazma basin, the fertile Suzdal *Opolye*, Tver, Yaroslavl, Kostroma and Kaluga, led by peasant clearance and monastic settlement networks around the Trinity-Sergius Lavra [2][23].

Under Ivan III (1462-1505AD) and Vasily III (1505-1533AD), Moscow annexed the Novgorod Republic (1478AD), the Grand Principality of Tver (1485AD), Vyatka (1489AD), the Upper Oka and Seversk principalities (1494-1503AD), Pskov (1510AD), Smolensk (1514AD) and Ryazan (1521AD) [43][44][45]. Because the Ilmen-Pskov arable heartland and the Finno-Karelian and Neva borderlands followed distinct institutional and military trajectories between 1478AD and 1721AD, the regional framework separates the Novgorod and Pskov lands (Region 19) from Olonets Karelia and Baltic Ingria (Region 20), while western Vyborg and Ladoga Kexholm are tracked in Region 4.

The Novgorod and Pskov lands (Region 19, 159,35k km^2) grew from 920k in 1400AD to 1,17M in 1500AD and 1,34M in 1550AD across the Shelonskaya, Derevskaya, Bezhetskaya and southern Vodskaya and Obonezhskaya *pyatiny* around Lake Ilmen, the Volkhov and the Lovat, together with the Pskov Republic (Pskov, Izborsk, Ostrov, Opochka and Gdov) and the southern border fortress of Velikie Luki [23][44]. Following the confiscation of Novgorodian boyar and archiepiscopal estates in 1478-1489AD and the settlement of Moscow cavalry *pomeshchiki*, the surviving *pistsovye knigi* of 1495-1505AD and the 1540s-1550sAD document a 14,5% demographic expansion between 1500AD and 1550AD, with Veliky Novgorod (~33k-35k in 1500-1550AD) and Pskov (~28k-30k inside the Okolny Gorod wall) ranking as the second and third cities of the Tsardom [12][23].

To the north, Olonets Karelia and Baltic Ingria (Region 20, 205,03k km^2) moved from 200k in 1400AD to 280k in 1550AD, 180k in 1615AD and 350k in 1720AD. This territory spanned the Neva and Gulf of Finland littoral (*Izhora/Ingria*, including Ivangorod founded in 1492AD opposite Narva, Yam, Koporye and Oreshek/Noteburg at the Neva outlet) together with Olonets-Onega Karelia (*Zaonezhskie* and *Lopatinskie pogosty*, with Olonets fortress founded in 1649AD) [23][39]. Unlike the Ilmen-Pskov grain belt, Region 20 combined slash-and-burn (*podseka*) agriculture, Ladoga-Onega fisheries, bog-iron smelting (*domnitsy*, and later the Petrine Olonets ironworks at Petrovskaya Sloboda in 1703AD) and the Neva-Ingrian fortresses ceded to Sweden under the Treaty of Stolbovo (1617AD) before Peter I's reconquest and the foundation of Saint Petersburg in 1703AD [12][39].

Along the western frontier, the Smolensk and Seversk borderlands (Region 21, 83,07k km^2, moving from 480k in 1400AD to 680k in 1550AD and 730k in 1720AD) guarded the Upper Dnieper and Desna-Bryansk corridor between Moscow and the Grand Duchy of Lithuania, anchored by Fyodor Kon's brick citadel at Smolensk (1595-1602AD), Dorogobuzh, Vyazma, Roslavl and Bryansk [1]. Across the Russian North, Pomorye, Vologda, the Northern Dvina and the Vychegda-Pechora Komi lands (Region 22, 1,39M km^2, moving from 210k in 1400AD to 340k in 1550AD, 480k in 1678AD and 420k in 1720AD) formed a *chernososhnye* (state-peasant) maritime, salt-boiling, fur-trading and Zyryan Komi region. After the English navigator Richard Chancellor reached the Northern Dvina in 1553AD, the White Sea trade route stimulated urban growth along the Sukhona-Dvina axis linking Vologda (13k in 1550AD), Veliky Ustyug (5k), Solvychegodsk (4k), Kholmogory (8k), Arkhangelsk (founded 1584AD), Kargopol, Kola and the Pechora portage outpost of Pustozersk (1499AD) [1][42]. Region 22 peaked at 480k in 1678AD before declining to 420k by 1720AD as Petrine military conscription, naval timber levies and the 1713-1718AD decrees diverting foreign maritime trade from Arkhangelsk to Saint Petersburg (Region 20) drew population out of the White Sea basin [7].

The Oprichnina and the Livonian War

The demographic expansion of the old Muscovite heartland reversed during the second half of the reign of Ivan IV (1547-1584AD), when four overlapping shocks struck Regions 18, 19, 20 and 21 between 1565AD

and 1583AD [12][39][46]. Twenty-five years of mobilisation during the Livonian War (1558-1583AD) and extraordinary wartime levies (*pyatinnye dengi*, *yam* post obligations and siege-artillery transport) exhausted the tax-paying peasantry (*tyaglye krestyane*) and *pomeshchik* service estates across the Novgorod, Pskov, Ingrian, Smolensk and Zamoskovny districts [12][39]. At the same time, Ivan IV's division of the realm into the *Zemshchina* and the *Oprichnina* (1565-1572AD) brought estate confiscations, heavy exactions and the punitive expedition against Tver, Torzhok, Klin and Veliky Novgorod in January-February 1570AD, killing thousands of inhabitants and disrupting the regional grain trade [1][12].

Environmental and military disasters compounded the fiscal crisis. Early frosts and harvest failure in 1569-1570AD triggered severe famine across central and north-western Russia, accompanied by a lethal plague epidemic (*morovoe povetrie*) throughout Regions 18-20 [12]. In May 1571AD, Khan Devlet I Giray bypassed the Oka defences (Region 23) and burned the wooden suburbs of Moscow (Region 18), killing tens of thousands and carrying off an estimated 60k-100k captives before Prince Mikhail Vorotynsky defeated the Crimean host at the Battle of Molodi in 1572AD [47][48].

The cadastral books compiled in the 1580sAD and early 1590sAD document widespread rural abandonment (*zapustenie*) [1][12]. In Novgorod and Pskov (Region 19), between 75% and 90% of previously cultivated *obzhi* were registered as deserted wasteland (*pustoshi* or *porozhnie zemli*); regional population dropped by 28,4%, from 1,34M in 1550AD to 960k in 1580AD, while Olonets Karelia and Baltic Ingria (Region 20) fell by 21,4% (280k to 220k) during Pontus De la Gardie's Swedish capture of Ivangorod, Yam and Koporye (1581-1590AD, recovered by Russia at the 1595AD Truce of Teusina) [12][39]. The Moscow and Zamoskovny core (Region 18) contracted from 3,05M (1550AD) to 2,78M (1580AD), and Smolensk and Seversk (Region 21) declined from 680k to 640k [1][45]. Conversely, fugitive peasants moved north into Pomorye (Region 22, 340k to 380k), south into the Belgorod and Sevsk Razryads (Region 24, 290k to 340k) and east into Kazan and the Middle Volga (Region 25, 940k to 960k) and Perm-Urals (Region 26, 330k to 360k), prompting the Muscovite crown to introduce the *zapovednye leta* ("forbidden years", from 1581AD) suspending peasant departure on St George's Day (*Yuryev den*) [19][46].

Belarus and Ukraine Before 1648AD

The Belarusian Voivodeships

In contrast to late 16th-century Muscovy, the four Belarusian regional clusters of the Grand Duchy of Lithuania (Regions 5-8, covering 246,65k km^2) experienced sustained demographic and urban expansion between 1400AD and 1648AD, with the combined Belarusian population rising from 950k in 1400AD to 1,3M in 1500AD, 1,6M in 1550AD, 1,88M in 1600AD and 2,5M on the eve of the mid-17th-century wars in 1648AD [9][13][26]. In the Polotsk and Vitebsk Voivodeships (Region 5, 32,58k km^2), anchored by the Western Dvina export artery to Riga and the commercial cities of Polotsk (~20k in 1648AD), Vitebsk (~17k), Drissa and Lepel, the population expanded from 240k in 1400AD to 390k in 1550AD, dipped temporarily to 350k in 1580AD during Ivan IV's occupation of Polotsk (1563-1579AD) and Stephen Bathory's reconquest, and climbed to 560k in 1648AD [13][49]. Across the Grodno and Novogrudok Voivodeships (Region 6, 48,97k km^2), comprising Black Ruthenia (*Chornaya Rus*) and the fertile Ponemanje basin around Grodno (~13,5k), Novogrudok (~9k), Slonim (~7,2k), Lida (~6k) and Volkovysk (~5,5k), the population grew from 280k in 1400AD to 470k in 1550AD and 740k in 1648AD [13][26].

Further east, the Minsk Voivodeship and the Orsha-Mogilev Dnieper corridor (Region 7, 102,28k km^2) spanned the central Belarusian watershed and the Upper Dnieper, Berezina and Sozh river arteries. Its population rose from 260k in 1400AD to 430k in 1550AD and 680k in 1648AD, led by Mogilev (~25k in 1648AD, the largest city in Belarus and the principal transit market between Moscow and Poland-Lithuania), Minsk (~13k), Slutsk (~12,5k), Orsha (~9,5k), Mstislavl (~9k), the Radziwill residence at Nesvizh (~7,5k), Gomel and Bobruysk [9][49]. In the south-west, the Brest-Litovsk Voivodeship and

Pinsk-Turov-Mozyr Polesia (Region 8, 62,82k km^2) encompassed the Western Bug gateway at Brest-Litovsk (~15k in 1648AD, granted Magdeburg Law in 1390AD and site of the 1596AD Church Union) alongside the Pripyat river-marshland towns of Pinsk (~12k), Mozyr (~6k) and Turov, with its population tripling from 170k in 1400AD to 310k in 1550AD and 520k in 1648AD [9][13].

This growth rested on Queen Bona Sforza's and Sigismund II Augustus's Voloka Reform (*Ustava na voloki*, 1557AD), which consolidated scattered peasant *dvorishcha* into regular three-field linear villages divided into standard 21,36 ha *voloka* plots, paired with the chartering of more than 60 Magdeburg-law municipalities and over 300 noble market towns (*miasteczka*) [13][49].

Crown Ukraine and the Left Bank

Across the Ukrainian and Crimean territories (Regions 9-17, covering 607,83k km^2), the total population rose from 2,51M in 1400AD to 3,16M in 1500AD, 3,71M in 1550AD, 4,66M in 1600AD and 6,1M in 1648AD [5][27]. Following the Union of Lublin (1569AD), which transferred Volhynia, Podolia-Bratslav and the Kyiv Voivodeship from Lithuania to the Polish Crown, settlement advanced steadily from the western palatinates into the Dnieper borderlands.

The Ruthenian and Belz Voivodeships of Galicia, together with Moldavian Bukovina and Hungarian Carpathian Rus (Region 9, 71,56k km^2), formed the most densely settled East Slavic territory under Polish Crown rule, expanding from 820k in 1400AD to 1,18M in 1550AD and 1,75M in 1648AD (24,5 persons/km^2). Its metropolitan capital Lviv (~25k in 1648AD) linked Black Sea and Levantine trade with Central Europe, flanked by Drohobych (~7,2k), Brody (~6,5k), Halych, Ternopil and Zhovkva [5][27]. To the north-east, the Volhynian Voivodeship (Region 10, 45,18k km^2), seat of the great Ruthenian princely houses of Ostrogski, Zaslawski, Zbaraski, Sanguszko and Czartoryski, grew from 420k in 1400AD to 620k in 1550AD and 920k in 1648AD, anchored by Ostroh (~12k, site of the Ostroh Academy and the 1581AD Slavonic Bible), Lutsk (~11k), Kremenets (~8k), Dubno (~7,5k) and Volodymyr (~6k) [5]. Across the Dniester and Southern Bug forest-steppe uplands, the Podolian and Bratslav Voivodeships (Region 11, 39,62k km^2) rose from 350k in 1400AD to 540k in 1550AD and 910k in 1648AD, led by the royal cliff fortress of Kamianets-Podilskyi (~15k), Vinnytsia (~9k), Bratslav (~8,5k), Bar (~7,5k) and Medzhybizh (~7k) [5][27].

Further east, the Right-Bank Kyiv Voivodeship and Cossack regiments (Region 12, 56,76k km^2), covering the Right-Bank Dnieper, Teterev, Irpin, Ros and Tyasmin basins, surged from 280k in 1400AD and 440k in 1550AD to 920k in 1648AD as magnate *slobody* (granting 15-30 years of tax exemption) and Registered Cossack regiments proliferated around Kyiv (~18k in 1648AD, intellectual seat of Metropolitan Petro Mohyla's Collegium), Uman (~11k), Bila Tserkva (~10,5k), Korsun (~9,5k) and Chyhyryn (~8k, Bohdan Khmelnytsky's residence and first Hetmanate capital) [5][27]. Beyond the Dnieper, in the Northern Left-Bank Hetmanate (Region 13, 55,78k km^2, covering Chernihiv, Nizhyn, Starodub, Putyvl, Baturyn and Hlukhiv) and the Southern Left-Bank Hetmanate (Region 14, 52,08k km^2, covering Pereyaslav, Lubny, Myrhorod, Pryluky, Poltava and Hadiach), the 1400AD populations stood at 210k and 110k respectively. After the 1618AD Truce of Deulino transferred Chernihiv-Seversk to Poland and the trans-Dnieper Wisniowiecki latifundium (*Wisniowiecczyzna*) centred on Lubny expanded from ~600 households in 1600AD to over 38k households (~230k inhabitants across 56 trans-Dnieper towns) by 1645AD, Region 13 reached 490k and Region 14 reached 460k in 1648AD (950k combined across the Left Bank) [5][27].

Seventeenth-Century Crises

The Time of Troubles

Following partial demographic recovery under Tsar Feodor Ivanovich and Boris Godunov (1584-1600AD) - which lifted Zamoskovny Moscow (Region 18) to 2,95M and Novgorod-Pskov (Region 19) to 1,04M in 1600AD - the Tsardom of Russia was struck by the Huaynaputina volcanic-winter famine of 1601-1603AD and the civil and international wars of the Time of Troubles (*Smuta*, 1604-1618AD) [1][50][51]. Post-*Smuta* inspection cadastres (*dozornye knigi* of 1614-1619AD) record steep contractions across central and north-western Russia between 1600AD and 1615AD [1][21].

The Moscow and Zamoskovny core (Region 18) fell by 22,7%, from 2,95M in 1600AD to 2,28M in 1615AD, amid famine, the Bolotnikov rebellion, the Tushino camp, the 1610-1612AD Polish garrison in Moscow and *Lisowczycy* light-cavalry raids. The Novgorod and Pskov Five Pyatiny (Region 19) plunged by 31,7% (and by 47% relative to 1550AD), from 1,04M in 1600AD to 710k in 1615AD, during Jacob De la Gardie's Swedish occupation of Veliky Novgorod (1611-1617AD) and Gustavus Adolphus's siege of Pskov (1615AD). Olonets Karelia and Baltic Ingria (Region 20) dropped by 25%, from 240k in 1600AD to 180k in 1615AD, mirroring Vyborg and Kexholm (Region 4, 95k to 80k). When the Treaty of Stolbovo (1617AD) ceded Baltic Ingria (*Ivangorod, Yam, Koporye and Noteburg*) alongside Kexholm County to Sweden, tens of thousands of Orthodox Karelians and Izhorians fled Swedish rule in the 1620s-1650sAD into the Bezhetskiy Verkh, Tver, Torzhok and Tikhvin *uezds* (Regions 18-19) [38][40][41].

Along the western and southern approaches, Smolensk and Seversk (Region 21) fell by 26,1% (690k to 510k) following the twenty-month Polish siege of Smolensk (1609-1611AD) and its cession at Deulino (1618AD), while Ryazan, Tula and Oryol (Region 23) declined from 1,02M to 820k and the Belgorod and Sevsk Razryads (Region 24) dipped from 410k to 380k [51][52]. Conversely, sheltered Pomorye and Vologda (Region 22, 410k to 420k), Vyatka, Perm and the Urals (Region 26, 390k to 405k) and the eight Siberian zones (Regions 31-38, 357k to 375k) escaped devastation and financed the 1611-1612AD Second Volunteer Army of Minin and Pozharsky from Nizhny Novgorod (Region 25) and Yaroslavl (Region 18) [1].

The Deluge in Lithuania and Belarus

During the Russo-Polish War of 1654-1667AD (*Potop* or *Krovavy Potop*), both Core Lithuania (Region 3, falling from 1,06M to 640k, or -39,6%) and the four Belarusian regions (Regions 5-8) suffered one of the steepest demographic contractions in early modern Europe [9][13]. Comparing the 1648-1650AD and 1667AD *podymne* hearth-tax registers across all Grand Duchy *powiaty*, J. Morzy (1965) and A. M. Karpachev and P. G. Kozlovsky (1972) showed that Belarus lost 48,8% of its total population, falling from 2,5M in 1648AD to 1,28M at the Truce of Andrusovo in 1667AD.

Within Belarus, the Polotsk and Vitebsk Voivodeships (Region 5) fell by 51,8% (560k to 270k) amid Dvina sieges, the 1657-1658AD plague and forced resettlements of skilled Belarusian townspeople (*belorustsy*) to Moscow [13]. The Minsk, Orsha and Mogilev Dnieper corridor (Region 7) dropped by 51,5% (680k to 330k) following the sack of Mstislavl (1654AD), the 1661AD Mogilev uprising and continuous campaigning along the Smolensk-Orsha-Minsk road [9]. Brest-Litovsk and Pinsk-Turov-Mozyr Polesia (Region 8) declined by 50% (520k to 260k), while the Grodno and Novogrudok Voivodeships (Region 6) contracted by 43,2% (740k to 420k) [9][13].

Although the four Belarusian regions partially recovered to 1,45M in 1678AD and 1,63M in 1700AD, the Great Northern War (1700-1721AD) - marked by Russian, Swedish, Saxon and Polish operations across Grodno, Polotsk, Lesnaya (1708AD) and Mogilev (burned in 1708AD), alongside the 1708-1711AD plague - pushed Belarus back to 1,43M in 1710AD and 1,55M in 1720AD, still 38% below its 1648AD peak [9][16][49].

The Ruin in Ukraine and Sloboda Colonisation

In Ukraine, the Khmelnytsky Uprising (1648-1657AD), the 1654AD Pereyaslav Agreement and the thirty-year multi-sided conflict known as *The Ruin* (*Ruina*, 1657-1687AD) shifted the demographic centre of gravity from Right-Bank Ukraine (Regions 10, 11 and 12) to the Left-Bank Hetmanate (Regions 13 and 14) and Sloboda Ukraine (Region 15) [5][27][53][54]. Between 1648AD and 1678AD, the Right-Bank Kyiv Voivodeship and Cossack regiments (Region 12) lost 60,9% of their population, falling from 920k in 1648AD to 520k in 1667AD, 360k in 1678AD and 320k in 1700AD after the Ottoman-Tatar Chyhyryn campaigns (1674-1678AD, which destroyed Uman and Chyhyryn), Hetman Ivan Samoilovych's 1678-1679AD *Velykyi Zhin* (forced trans-Dnieper evacuation to the Left Bank) and the 1686AD Eternal Peace buffer-zone clause, while the city of Kyiv itself remained under Russian garrison rule on the Dnieper cliff at ~15k-18k [5][53]. Simultaneously, Podolia and Bratslav (Region 11, ruled by the Ottoman Empire as the Eyalet of Kamanice in 1672-1699AD) fell by 58,2% from 910k to 380k in 1678AD before recovering to 510k by 1720AD after the 1699AD Treaty of Karlowitz; Volhynia (Region 10) dropped from 920k to 580k (recovering to 680k in 1720AD); and Galicia-Belz (Region 9) contracted from 1,75M to 1,38M (reaching 1,62M in 1720AD) [5].

Absorbing waves of Right-Bank settlers, the Northern Left-Bank Hetmanate (Region 13, covering Chernihiv, Nizhyn, Starodub and Putyvl, plus the Hetman capitals of Baturyn in 1669-1708AD and Hlukhiv in 1708-1764AD) grew continuously from 490k in 1648AD to 560k in 1667AD, 620k in 1678AD and 790k in 1720AD (+61,2%) [53]. The Southern Left-Bank Hetmanate (Region 14, comprising the Pereyaslav, Lubny, Myrhorod, Pryluky, Hadiach and Poltava regiments) rose from 460k in 1648AD to 540k in 1667AD, 610k in 1678AD and 760k in 1720AD (+65,2%), weathering the 1708-1709AD Swedish invasion and the Battle of Poltava in June 1709AD [5][53]. Further east, Sloboda Ukraine (Region 15, 60,98k km^2, spanning the Kharkiv, Sumy, Okhtyrka, Ostrogozhsk and Izium regiments) was colonised by Ukrainian *cherkasy* migrants under Tsar Aleksey Mikhailovich's autonomous regimental charters (Ostrogozhsk in 1652AD, Sumy in 1652AD, Kharkiv in 1654AD, Okhtyrka in 1654AD and Izium in 1681AD), surging from 110k in 1600AD and 160k in 1648AD to 250k in 1667AD, 310k in 1678AD, 380k in 1700AD and 400k in 1720AD (+150% above 1648AD) [17][54][55].

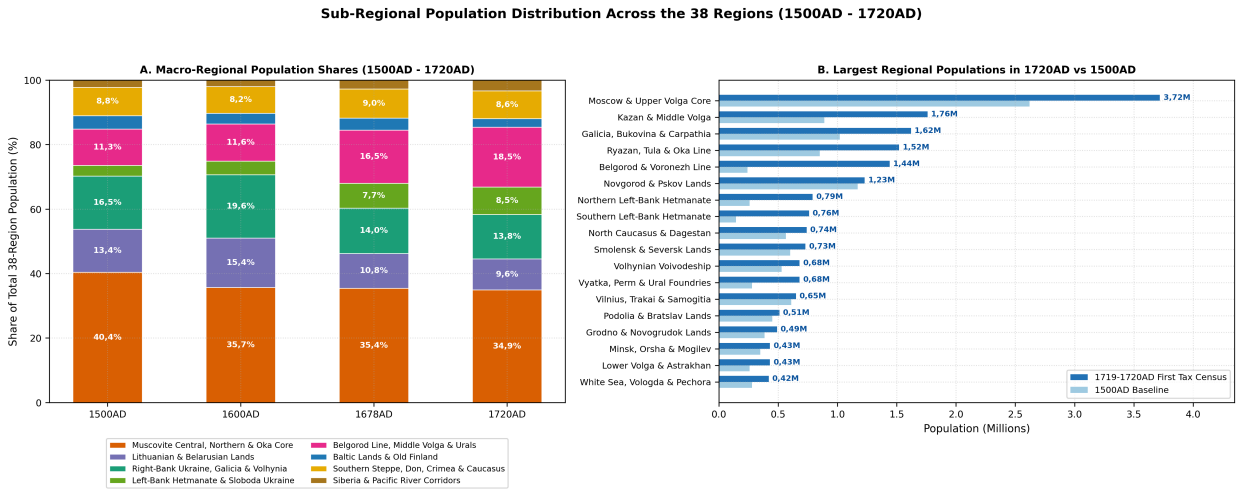


Figure 3. Percentage population losses and gains across key regions during major seventeenth- and early eighteenth-century crises (Panel A) and Russian versus native population growth across Siberia from 1580AD to 1720AD (Panel B).

The Southern Frontier and the Steppe

The Oka and Belgorod Lines

South of the Muscovite forest zone, the conversion of the black-earth forest-steppe from a Crimean-Nogai raiding ground (*Pole*) into the main granary of Russia unfolded across two frontier regions [18][52][56]. Along the Ryazan, Tula, Oryol and Great Oka Abatis Line (Region 23, 95,59k km^2), settlement was shielded by the felled-forest barriers, earthworks and artillery gates of the Great Abatis Line (*Bolshaya Zasechnaya Cherta*, built in 1521-1566AD and rebuilt in 1638AD) running from Kozelsk, Belev and Odoyev through Tula and Venev to Ryazan (*Pereyaslavl-Ryazansky*), Kolomna, Oryol (1566AD) and Yelets (1592AD). This region expanded from 680k in 1400AD to 980k in 1550AD, recovered from its 1615AD *Smuta* dip (820k) and reached 1,08M in 1648AD, 1,29M in 1678AD and 1,52M in 1720AD (15,9 persons/km^2), with Tula (19k in 1720AD, home to the 1632AD Vinius ironworks and the 1712AD Imperial Arms Plant) serving as Russia's leading weapons town [7][18].

Further south, in the Belgorod and Sevsk Razryads (Region 24, 148,78k km^2), the Military Chancellery (*Razryadny Prikaz*) blocked the Muravsky, Izyumsky and Kalmiussky Tatar trails across the Kursk, Belgorod, Voronezh, Lipetsk, Kozlov and Tambov black-earth districts by founding Voronezh (1586AD), Livny (1586AD), Stary Oskol (1593AD), Belgorod (1596AD) and Kursk (1596AD), and in 1635-1658AD built the continuous 800 km Belgorod Fortified Line (*Belgorodskaya Cherta*) extending to Kozlov (1635AD) and Tambov (1636AD) [47][56]. Organised into the Belgorod Razryad (1658AD) and Sevsk Razryad, Region 24 grew more than sevenfold from 190k in 1400AD and 290k in 1550AD to 620k in 1648AD, 980k in 1678AD and 1,44M in 1720AD (+132,3% above 1648AD) [7][57]. Its countryside was farmed by *odnodvortsy* ("single-homesteaders" descended from seventeenth-century petty gentry, cossacks, musketeers and dragoons, exceeding 450k male souls by the 1720sAD), alongside growing landlord serf estates and Peter I's Azov fleet shipyards at Voronezh (16k in 1720AD) and Lipetsk [18][58].

Frontier Region	Active Phase	Geographic Span	Main Towns & Forts	Main Social Groups	Population Growth
Great Oka Abatis Line (Region 23)	1521-1638AD	600 km (Kozelsk - Tula - Kolomna - Ryazan)	Tula, Ryazan, Oryol, Kolomna, Yelets, Belev	Pomeshchiki, streltsy, gunsmiths & landlords' serfs	680k (1400AD) to 1,08M (1648AD) to 1,52M (1720AD)
Belgorod & Sevsk Razryads (Region 24)	1586-1658AD	800 km (Belgorod - Voronezh - Kozlov - Tambov)	Belgorod, Voronezh, Kursk, Tambov, Kozlov, Stary Oskol	Odnodvortsy (deti boyarskie), streltsy & serfs	190k (1400AD) to 620k (1648AD) to 1,44M (1720AD)
Kazan & Simbirsk-Penza Line (Region 25)	1552-1683AD	650 km (Saransk - Karsun - Simbirsk - Penza - Syzran)	Kazan, Nizhny Novgorod, Simbirsk, Saransk, Penza, Samara	Service Tatars/ Mordvins, yasak Chuvash/Mari & serfs	780k (1400AD) to 1,18M (1648AD) to 1,76M (1720AD)
Sloboda Ukraine Regiments (Region 15)	1652-1681AD	530 km (Sumy - Okhtyrka - Kharkiv - Ostrogozhsk - Izium)	Kharkiv, Sumy, Okhtyrka, Izium, Ostrogozhsk, Valuyki	Ukrainian Cherkasy regimental cossacks & burghers	50k (1400AD) to 160k (1648AD) to 400k (1720AD)

N. & S. Left-Bank Hetmanate (Regions 13-14)	1648-1720AD	Desna, Sula, Psyol & Vorskla regiment network	Baturyn, Hlukhiv, Nizhyn, Chernihiv, Poltava, Lubny	Registered Cossacks, pospolyti peasants & burghers	320k (1400AD) to 950k (1648AD) to 1,55M (1720AD)
Bashkir Four Darugas & Yaik (Region 27)	1574-1720AD	Belaya (Agidel), Ufa & Yaik river frontier	Ufa (1574AD), Birsk (1663AD), Menzelinsk, Yaitskiy Gorodok	Votchinniki Bashkirs (4 darugas), teptyari & Yaik Cossacks	180k (1400AD) to 260k (1648AD) to 340k (1720AD)

Table 4. Fortified defence lines, frontier districts and Cossack regiments across Ukraine, the southern Russian forest-steppe and the Middle Volga-Ural borderlands from 1400AD to 1720AD.

Zaporozhia, Crimea, the Lower Volga, the Don and the North Caucasus

Beyond the settled forest-steppe lines lay five steppe, maritime and mountain regions. In the Zaporozhian Sich and Pontic Wild Fields (Region 16, 193,86k km^2, rising from 60k in 1400AD to 130k in 1648AD and 180k in 1720AD), settlement clustered around the island strongholds of the Zaporozhian Cossack Host below the Dnieper rapids (Khortytsia, Tomakivka, Bazavluk, Mykytyn Rih, Chortomlyk in 1652-1709AD, Kamianka and Oleshky), the Polish river fort of Kodak (1635AD), surrounding winter fishing-pastoral homesteads (*zymivnyky*), the Ottoman Black Sea fortress of Ochakiv (*Ozu*) and the Bakhmut salt-boiling forts [27][54]. To the south, the Crimean Khanate and Ottoman Eyalet of Kefe (Region 17, 32k km^2, growing from 210k in 1400AD to 360k in 1648AD and 370k in 1720AD) was divided after 1475AD between the direct Ottoman coastal province of Kefe (*Kaffa/Feodosia*) - holding ~65k-85k Muslim, Greek, Armenian and Jewish inhabitants in 16th-century Ottoman *tahrir defterleri* across Kaffa, Sudak, Balaklava, Mangup, Kerch and Taman [28][59] - and the Giray Crimean Khanate (~270k-290k inhabitants), anchored by Bakhchisarai (~20k-22k), Qarasubazar (*Karasubazar*, ~11k-12,5k) and Gozleve (*Yevpatoria*, ~10k) [28].

On the Caspian-Volga frontier, the Lower Volga garrison towns, Astrakhan and the Kalmyk Khanate (Region 28, 251,4k km^2) rose from 240k in 1400AD to 320k in 1648AD and 430k in 1720AD. Following Ivan IV's conquest of the Khanate of Astrakhan (1556AD), Muscovy secured the Volga-Caspian trade artery with Astrakhan (22k in 1720AD, hosting Russian, Tatar, Bukharan, Persian, Armenian and Indian merchant compounds), Tsaritsyn (1589AD) and Saratov (1590AD), while Oirat pastoralists under Kho Orluk and Ayuka Khan (1669-1724AD) established the Kalmyk Khanate (~40k-50k *kibitki* or ~200k-250k nomads) across the surrounding Caspian-Volga steppes [7][48][60]. West of the Volga bend, the Don Cossack Host and Azov (Region 29, 169,53k km^2) expanded from 90k in 1400AD to 165k in 1648AD, 240k in 1700AD, 230k in 1710AD and 250k in 1720AD. Along the Lower and Middle Don, Seversky Donets, Khopyor and Medveditsa, the autonomous Don Cossack Host established over 120 fortified *gorodki* governed from Razdory and Cherkassk (capital from 1644AD, ~8,5k in 1720AD, plus Mikhaylovskiy Gorodok on the Khopyor), contesting Ottoman Azov (held by the Cossacks in 1637-1642AD and by Peter I in 1696-1711AD alongside newly built Taganrog in 1698AD) and experiencing a brief demographic dip during the Bulavin Rebellion of 1707-1708AD [47][61].

South of the steppe, the North Caucasus (Region 30, 263,21k km^2, moving from 520k in 1400AD to 665k in 1648AD and 740k in 1720AD) encompassed Adyghe and Circassian societies, Greater and Lesser Kabarda (allied with Moscow from 1557AD), Vainakh (*Chechen* and *Ingush*) highland tower communities and piedmont settlements [62], the Russian Terek delta fortress of Terskiy Gorod (*Terki*, 1588AD) with the

Grebensky Cossacks, and the polities of Dagestan (Derbent, the Kumyk Shamkhalate of Tarki, Endirey, Kumukh, the Avar Khanate and mountain *jamaat* unions) [63].

The Volga, the Urals and Siberia

Kazan, Perm and Bashkiria

East of the Oka and Sura rivers, the conquest of the Khanate of Kazan (1552AD) integrated the Middle Volga, Kama and Ural basins into the Russian state across three regions. In the former Khanate of Kazan, Nizhny Novgorod, the Chuvash-Mari lands and the Simbirsk-Penza Line (Region 25, 237,58k km^2), the population grew from 780k in 1400AD to 940k in 1550AD, 1,18M in 1648AD and 1,76M in 1720AD. This territory combined Nizhny Novgorod (15k in 1720AD, adjacent to the Makaryev Fair), Arzamas, Kasimov and Temnikov with the former Kazan Khanate (Kazan at 30k in 1720AD, Sviyazhsk founded in 1551AD, Cheboksary in 1555AD and Samara in 1586AD) and the Simbirsk-Kama (1648-1656AD) and Penza-Syzran (1663-1683AD) fortification lines (Saransk in 1641AD, Simbirsk in 1648AD, Penza in 1663AD and Syzran in 1683AD), populated by Russian peasants alongside *yasak* and service (*sluzhilye*) Volga Tatars, Chuvash, Mordvins, Mari and Udmurts [7][64][65][66].

To the north-east, Great Perm, Vyatka, the Solikamsk-Stroganov salt lands and the early Demidov Ural foundries (Region 26, 479,79k km^2) expanded from 220k in 1400AD to 330k in 1550AD, 480k in 1648AD and 680k in 1720AD, linking the *chernososhnye* grain basin of Vyatka (*Khlynov*) with Perm Velikaya (Cherdyn, Solikamsk at 9,8k in 1720AD and the Stroganov Kama-Chusovaya salt domains), the Sylva agricultural hub of Kungur (1648/1663AD) and Nikita Demidov's Petrine blast-furnace ironworks in the Middle Urals (Nevyansk in 1701AD, Kamensk in 1701AD and Alapaevsk in 1704AD) [7][8]. South of Perm, the Bashkir Four Darugas and the Yaik Cossack steppe frontier (Region 27, 484,83k km^2, rising from 180k in 1400AD to 215k in 1550AD, 260k in 1648AD and 340k in 1720AD) were administered after Ufa's foundation (1574AD) through the four Bashkir *darugas* (Kazan, Nogai, Osinskaya and Siberian), where hereditary Bashkir *votchinniki* clans defended their patrimonial land rights in the uprisings of 1662-1664AD, 1681-1684AD and 1704-1711AD alongside Russian fortresses (Birsk in 1663AD and Menzelinsk), *teptyari* migrants and the Yaik Cossack Host at Yaitskiy Gorodok [7][57].

Siberian Colonisation

East of the Ural divide, Russian servicemen (*sluzhilye lyudi*), fur trappers (*promyshlenniki*) and state plough peasants (*pashennye krestyane*) advanced 6k km from Yermak's 1582AD capture of Qashliq (*Isker*) to Okhotsk (1647AD), the Bering Strait (1648AD), the Amur (1651-1689AD) and Kamchatka (1697AD) [6][20][67][68]. Because early modern Siberian settlement followed east-west river-portage highways (*voloki*), southern fortified ploughlands (*gosudareva desyatinnaya pashnya*), nomadic steppe borderlands and northern *yasak* fur districts, Northern Asia (12,81M km^2 across Regions 31-38, rising from 300k in 1400AD and 338k in 1580AD to 435k in 1648AD, 545k in 1678AD and 760k in 1720AD) is divided into eight regional zones (Panel B of Figure 1, Panel B of Figure 3 and Table 5).

Along the Pomor-Mangazeya maritime and Taz-Turukhansk Arctic portage route (Region 31, 1,76M km^2), Pomor mariners sailing *kochi* across the Kara Sea and the Yamal portage founded Obdorsk (1595AD), Mangazeya (1601AD) on the Taz River and Turukhansk (1607AD), lifting seasonal and indigenous Nenets, Enets, Nganasan and Selkup numbers from 16k in 1400AD and 19k in 1580AD to a peak of 28k in 1615AD and 26k in 1648AD, before Tsar Mikhail Romanov's 1619AD decree closing the Mangazeya Sea Route, sable depletion and the 1642AD Mangazeya fire relocated administration to Novaya Mangazeya (*Turukhansk*, 1672AD) and reduced Region 31 to 18k by 1720AD [6][67][69]. South of the Arctic circle, the Tobolsk-Irtysh-Middle Ob river portage highway (Region 32, 871,34k km^2, growing from 42k in 1400AD and 49k in 1580AD to 88k in 1648AD, 118k in 1678AD and 158k in 1720AD) was controlled via the Babylka

Ural Portage (1598AD) at Verkhoturye (1598AD), Turinsk (1600AD), Pelym (1593AD), Tyumen (1586AD) and the Siberian capital Tobolsk (1587AD, 15k in 1720AD), extending down the Lower Irtysh (Berezov in 1593AD) and Middle Ob (Surgut in 1594AD, Narym in 1596AD and Ketsky Ostrog in 1602AD) across Khanty, Mansi and Siberian Tatar *yasak* volosts [7][70][71][72].

The southern Tobol-Ishim-Tara-Tomsk-Kuznetsk fortified agricultural frontier (Region 33, 359,19k km^2, rising from 38k in 1400AD and 47k in 1580AD to 86k in 1648AD, 145k in 1678AD and 254k in 1720AD) stretched along the forest-steppe chernozem from the Iset-Tobol *slobody* (Shadrinsk in 1662AD and Kurgan in 1679AD), Tara (1594AD) and Omsk (1716AD) to Tomsk (1604AD, 8,5k in 1720AD) and Kuznetsk (1618AD). Region 33 became the main demographic engine of Siberia, expanding more than five-fold between 1580AD and 1720AD and housing over 60% of Northern Asia's Russian plough peasantry [20][73][74]. Immediately to its south, the nomadic Baraba-Altai-Minusinsk-Tuva steppe borderlands (Region 34, 821,19k km^2, moving from 74k in 1400AD and 81k in 1580AD to 84k in 1648AD, 80k in 1678AD and 100k in 1720AD) remained a contested zone of dual tribute (*dvoedanstvo*) inhabited by Baraba Tatars, Teleuts, Shors, Tuvans and the four Yenisei Kirghiz *ulusy* of Minusinsk until the 1703AD Dzungar relocation of the Kirghiz elite enabled Russian construction of Abakansky Ostrog (1707AD) and Biysky Ostrog (1709AD) [6][73].

In Central and Eastern Siberia, the Ket-Yeniseysk-Angara-Ilimsk-Irkutsk river portage spine (Region 35, 1,93M km^2, growing from 28k in 1400AD and 32k in 1580AD to 44k in 1648AD, 66k in 1678AD and 98k in 1720AD) linked the Makovskiy Ket-Yenisei portage (1618AD) through Yeniseysk (1619AD, 8k in 1720AD) and Krasnoyarsk (1628AD) up the Angara to Ilimsk (1630AD, on the Lena Portage) and Irkutsk (1661AD, 6,5k in 1720AD) among Western Buryats, Kets and Evenks [6][7][67]. Beyond Lake Baikal, Transbaikal Dauria and the Upper Amur frontier (Region 36, 1,75M km^2, rising from 34k in 1400AD and 38k in 1580AD to 40k in 1648AD, 48k in 1678AD and 62k in 1720AD) was anchored by Nerchinsk (1658AD), Selenginsk (1665AD) and Udinsk (1666AD) among Khori-Selenga Buryats, Evenks and Daurs, after Qing sieges of Albazin (1651-1689AD) and the Treaty of Nerchinsk (1689AD) redirected Russian settlement into Nerchinsk silver-lead mining (1704AD) and Kiakhta-Selenga caravan trade [6][75][76].

To the north-east, the central Lena-Vilyuy-Aldan basin (Region 37, 1,37M km^2, moving from 38k in 1400AD to 42k in 1580AD and 1648AD, 44k in 1678AD and 47k in 1720AD) centred on Yakutsk (1632AD), Vilyuysk (1634AD) and Olekminsk (1635AD) in the thermokarst *alaas* meadows of the Yakut (Sakha) cattle- and horse-breeding pastoralists (~28,5k-35k Sakha across 35 *yasak* volosts plus Evenks and ~4k-5k Russians by 1720AD) [6][77]. Finally, across the Northeast Arctic and Pacific *yasak* frontier (Region 38, 3,92M km^2, declining from 30k in 1400AD and 1580AD to 25k in 1648AD and 1678AD and 23k in 1720AD), spanning Okhotsk (1647AD), Anadyrsk (1649AD) and Nizhnekamchatsk (1697/1703AD), virgin-soil smallpox epidemics (1651-1653AD, 1669AD and 1691-1694AD) and frontier conflict reduced Yukaghir, Itelmen, Koryak, Even and Chukchi populations from 30k to 23k [6][76][78].

Siberian Zone (Reg. 31-38)	Area	Principal Ostrogi, Portages & Indigenous Ulusy	1580AD	1615AD	1648AD	1678AD	1720AD
31. Pomor-Mangazeya Maritime & Taz Portage	1,76M km^2	Obdorsk (1595AD), Mangazeya (1601AD; closed 1619AD), Turukhansk (1607/1672AD)	19k	28k	26k	19k	18k

32. Tobolsk-Irtysh-Middle Ob Portage Highway	871,34k km^2	Tyumen (1586AD), Tobolsk (1587AD), Berezov (1593AD), Surgut (1594AD), Verkhoturye (1598AD)	49k	64k	88k	118k	158k
33. S. Tobol-Tara-Tomsk-Kuznetsk Ploughland	359,19k km^2	Tara (1594AD), Tomsk (1604AD), Kuznetsk (1618AD), Shadrinsk (1662AD), Omsk (1716AD)	47k	60k	86k	145k	254k
34. Nomadic Baraba-Altai-Minusinsk-Tuva Steppe	821,19k km^2	Baraba & Teleut ulusy, Hongorai Kirghiz; Abakansky (1707AD), Biysky (1709AD)	81k	82k	84k	80k	100k
35. Ket-Yeniseysk-Angara-Ilimsk-Irkutsk Spine	1,93M km^2	Makovskiy (1618AD), Yeniseysk (1619AD), Krasnoyarsk (1628AD), Ilimsk (1630AD), Irkutsk (1661AD)	32k	34k	44k	66k	98k
36. Transbaikal Dauria & Upper Amur Frontier	1,75M km^2	Albazin (1651-1689AD), Nerchinsk (1658AD), Selenginsk (1665AD), Udinsk (1666AD)	38k	39k	40k	48k	62k
37. Central Lena-Vilyuy-Aldan (Yakutsk Alaas)	1,37M km^2	Yakutsk (1632AD), Vilyuysk (1634AD), Olekminsk (1635AD), Amginskaya Sloboda (1652AD)	42k	41k	42k	44k	47k
38. NE Arctic & Pacific Yasak (Kolyma-Kamchatka)	3,92M km^2	Verkhoyansk (1638AD), Srednekolymsk (1643AD), Okhotsk (1647AD), Anadyrsk (1649AD), Nizhnekamchatsk (1697AD)	30k	27k	25k	25k	23k

Total Siberia & Far East (Regions 31-38)	12,81M km^2	Over 50 ostrogi, slobody, portage stations and goroda across Northern Asia	338k (2k RU)	375k (25k RU)	435k (85k RU)	545k (165k RU)	760k (365k RU)

Table 5. Land areas, principal river fortresses and population totals across the eight Siberian zones from Yermak's crossing in 1580AD to the First Soul Revision of 1720AD.

Peter the Great's Reforms and the Main Towns

The reign of Peter I (1682/1689-1725AD) and the Great Northern War (1700-1721AD) reshaped both the provincial map and the leading towns of the realm [7][8][24]. After taking Swedish Noteburg (Shlisselburg) and Nyenskans on the Neva in Region 20, Peter I founded Saint Petersburg (May 1703AD) and the naval base of Kronstadt (1704AD), backed in 1710AD by the capture of Vyborg and Kexholm (Region 4) alongside Riga, Pernau, Arensburg and Reval (Region 1). Built by annual drafts of 20k-40k conscripted labourers (*rabotnye lyudi*) and populated by the compulsory move of noble households, foreign-trade merchants and government colleges (1712-1718AD), Saint Petersburg grew from ~5k in 1703AD to ~18k in 1710AD and 40k in 1720AD, lifting Region 20 from 270k in 1700AD to 350k in 1720AD and making the new Baltic port the second largest city in the Empire behind Moscow (Region 18, 160k) [7][24].

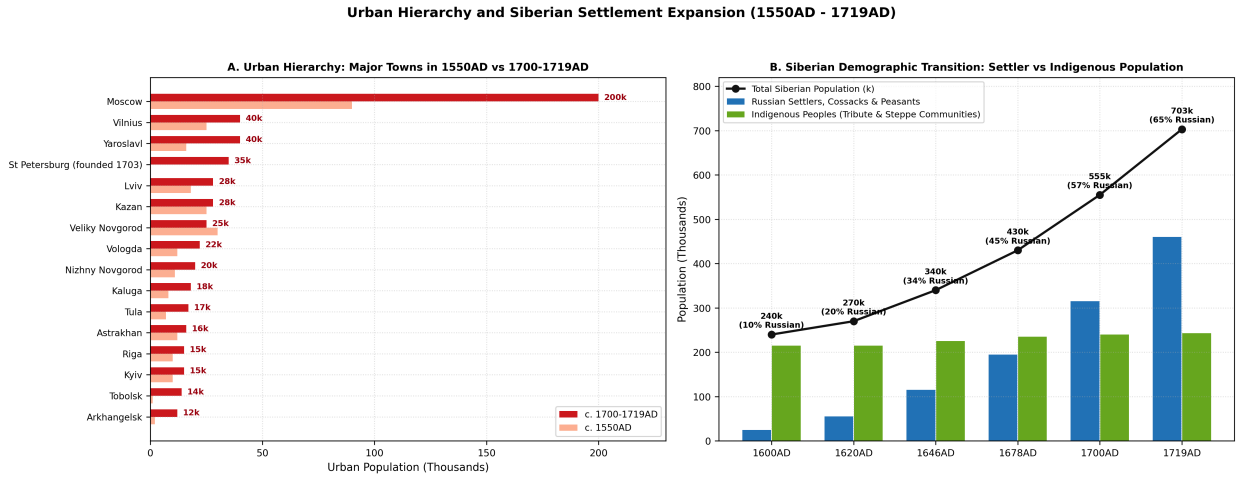


Figure 4. Population of the largest cities in 1500AD and 1648AD before the mid-century wars (Panel A) and the leading cities of the realm at the First Soul Revision in 1720AD after the rise of Saint Petersburg (Panel B).

Following the 1708-1710AD governorate (*guberniya*) reform (creating Moscow, Ingermanland/Saint Petersburg, Kiev, Smolensk, Arkhangelogorod, Kazan, Azov/Voronezh and Siberia, joined after 1710-1713AD by Riga/Livland and Reval/Estland) and the 1710AD household count - which revealed heavy wartime recruitment, plague deaths in the western provinces and widespread hiding of households (*utayka dvorov*) - the Senate ordered the First Soul Revision (1718-1721AD) [7][8][22]. By 1720AD, the main towns across the 38 regions (Figure 4) fell into three geographical groups.

Across Russia and Siberia (Regions 18-38, 14,37M total), behind Moscow (Region 18, 160k) and Saint

Petersburg (Region 20, 40k), the largest towns were Kazan (Region 25, 30k), Yaroslavl (Region 18, 26k), Astrakhan (Region 28, 22k), Kaluga (Region 18, 20k), Tula (Region 23, 19k), Voronezh (Region 24, 16k), Nizhny Novgorod (Region 25, 15k), Tobolsk (Region 32, 15k), Pskov (Region 19, 15k), Veliky Novgorod (Region 19, 14,5k), Kostroma (Region 18, 14,5k), Vologda (Region 22, 14k), Smolensk (Region 21, 14k), Kursk (Region 24, 12,5k), Kolomna (Region 23, 12,5k), Belgorod (Region 24, 11k) and Oryol (Region 23, 11k) [7][24].

In Ukraine and Crimea (Regions 9-17, 5,65M total), Lviv (Region 9, 25k), Bakhchisarai (Region 17, 20k) and Kaffa (Region 17, 19k) led the west and south, while Kyiv (Region 12, 18k), Nizhyn (Region 13, 14k), Chernihiv (Region 13, 13,5k), Hlukhiv (Region 13, 12,5k), Poltava (Region 14, 12k), Starodub (Region 13, 11,5k), Pereyaslav (Region 14, 11,5k), Kamianets-Podilskyi (Region 11, 11,5k) and Kharkiv (Region 15, 11k) were the chief towns of the Hetmanate, Sloboda Ukraine and Podolia [5][28][53].

In the Baltic provinces, Courland, Lithuania, Belarus and Old Finland (Regions 1-8, 2,81M total), which were still rebuilding after the 1708-1711AD plague and the sieges of the Great Northern War, Mogilev (Region 7, 18k) and Vilnius (Region 3, 13,5k, down from 28k in 1648AD) led the Lithuanian-Belarusian interior alongside Slutsk (Region 7, 9,5k), Polotsk (Region 5, 9k), Vitebsk (Region 5, 8,5k), Grodno (Region 6, 8,5k), Minsk (Region 7, 8k), Brest-Litovsk (Region 8, 8k) and Kaunas (Region 3, 6,5k), while Riga (Region 1, 9,5k), Reval/Tallinn (Region 1, 6,8k), Mitau/Jelgava (Region 2, 4,2k), Narva (Region 1, 4,2k), Vyborg (Region 4, 3,5k), Libau (Region 2, 3,2k) and Dorpat/Tartu (Region 1, 2,8k) were the chief Baltic ports and garrison towns [3][9][10][35].

Conclusion

Over the three centuries between 1400AD and the Treaty of Nystad in 1721AD, the population across the 38 historical regions of Russia, Ukraine, Belarus, Lithuania, the Baltic provinces and Siberia doubled from 11,36M to 22,83M. Yet that growth was far from even. In the western palatinates of Lithuania, Belarus and Right-Bank Ukraine, as well as in the Baltic provinces, steady fifteenth- and sixteenth-century expansion was undone by the mid-seventeenth-century Deluge, *The Ruin*, the famine of 1695-1697AD and the Great Plague of 1708-1711AD, leaving many western provinces with fewer people in 1720AD than they had held in 1648AD. By contrast, behind the timber and earth ramparts of the Oka, Belgorod and Simbirsk defence lines, and along the river highways of Left-Bank Ukraine, the Middle Volga, the Urals and Siberia, peasant and Cossack settlement turned open forest-steppe and taiga into the new demographic heartland of the Russian state.

When Peter I took the title of Emperor (*Imperator Vserossiyskiy*) at Nystad in September 1721AD, the 22,83M inhabitants of the 38 regions in 1720AD linked directly into the 23,89M baseline of the Early Imperial Russia series (1721-1815AD) [8][57]. Across the Baltic and western borderlands, Region 1 (300k in 1720AD) divided into Estland (105k in 1721AD) and Livland (195k); Region 2 (200k) continued as Courland (200k); Region 3 (650k) aligned with Vilna and Samogitia (655k); and Region 4 (110k) became Old Finland/Vyborg (110k). The additional 1,06M inhabitants between the 1720AD total (22,83M) and the 1721AD Early Imperial total (23,89M) come from two sources: 590k from bringing the Grand Duchy of Finland (390k in 1721AD) and Bessarabia (200k in 1721AD) into the 1721-1815AD territorial frame, and 468k (+2,05%) from post-war recovery and the follow-up military revision audits of 1721-1724AD [7][8].

Sources

1. Vodarsky, Yaroslav Evgenevich. *Naselenie Rossii Za 400 Let (XVI - Nachalo XX Vv.) [the Population of Russia over 400 Years, 16th to Early 20th Centuries]*. Prosveshchenie, 1973, https://search.worldcat.org/search?q=Population+Russia+over+400+Years+16th+Early+20th+Vodarsky+1973.

2. Kopanev, Aleksandr Ilich. “Naselenie Russkogo Gosudarstva v XVI v. [The Population of the Russian State in the 16th Century].” *Istoricheskie Zapiski*, vol. 64, 1959, pp. 233-54, https://cyberleninka.ru/article/n/naselenie-russkogo-gosudarstva-v-xvi-v.
3. Palli, Heldur. *Eesti Rahvastiku Ajalugu Kuni 1712. Aastani [Population History of Estonia up to 1712]*. Teaduste Akadeemia Kirjastus, 1996, https://search.worldcat.org/search?q=Population+History+Estonia+up+1712+Palli+1996.
4. Ochmanski, Jerzy. *Historia Litwy [History and Demographic Reconstruction of the Grand Duchy of Lithuania]*. Ossolineum, 1982, https://search.worldcat.org/search?q=History+Demographic+Reconstruction+Grand+Duchy+Lithuania+Ochmanski+1982.
5. Yakovenko, Nataliya Mykolaivna. *Narys Istorii Serednovichnoi Ta Rannomodernoi Ukrainy [Outline History of Medieval and Early Modern Ukraine]*. Krytyka, 2006, https://search.worldcat.org/search?q=Outline+History+Medieval+Early+Modern+Ukraine+Yakovenko+2006.
6. Dolgikh, Boris Osipovich. *Rodovoy i Plemennoy Sostav Narodov Sibiri v XVII Veke [Clan and Tribal Composition of the Peoples of Siberia in the 17th Century]*. Izdatelstvo Akademii Nauk SSSR, 1960, https://search.worldcat.org/search?q=Clan+Tribal+Composition+Peoples+Siberia+17th+Century+Dolgikh+1960.
7. Vodarsky, Yaroslav Evgenevich. *Naselenie Rossii v Kontse XVII - Nachale XVIII Veka (Chislennost, Soslovno-Klassovyy Sostav, Razmeshchenie) [the Population of Russia in the Late 17th and Early 18th Centuries: Size, Social Composition and Distribution]*. Nauka, 1977, https://doi.org/10.2307/1853555.
8. Kabuzan, Vladimir Maksimovich. *Narodonaselenie Rossii v XVIII - Pervoy Polovine XIX v. (Po Materialam Reviziy) [Population of Russia in the 18th and First Half of the 19th Century According to the Revisions]*. Izdatelstvo Akademii Nauk SSSR, 1963, https://search.worldcat.org/search?q=Population+Russia+18th+First+Half+19th+Century+According+Kabuzan+1963.
9. Karpachev, Avram Moiseevich, and Pavel Grigorevich Kozlovsky. *Dinamika Chislennosti Naseleniya Belorussii Vo Vtoroy Polovine XVII - XVIII Vv. [Dynamics of the Population Size of Belarus in the Second Half of the 17th and 18th Centuries]*. Nauka i Tekhnika, 1972, pp. 81-94, https://search.worldcat.org/search?q=Dynamics+Population+Size+Belarus+Second+Half+17th+18th+Karpachev+1972.
10. Dunsdorfs, Edgars. *Latvijas Vesture 1600-1710 [History and Historical Demography of Latvia, Livonia, Courland and Latgale, 1600-1710]*. Daugava, 1962, https://search.worldcat.org/search?q=History+Historical+Demography+Latvia+Livonia+Courland+Latgale+1600+Dunsdorfs+1962.
11. Saloheimo, Veijo. *Pohjois-Karjalan Historia II: 1617-1721 [History and Population Resettlement of Kexholm County and Northern Karelia, 1617-1721]*. Karjalaisen Kulttuurin Edistamissaatio, 1976, https://search.worldcat.org/search?q=History+Population+Resettlement+Kexholm+County+Northern+Karelia+1617+Saloheimo+1976.
12. Shapiro, Aleksandr Lvovich. *Agrarnaya Istoriya Severo-Zapada Rossii XVI Veka: Novgorodskie Pyatiny [Agrarian History of North-Western Russia in the 16th Century: The Novgorodian Pyatiny]*. Nauka, 1974, https://search.worldcat.org/search?q=Agrarian+History+North+Western+Russia+16th+Century+Novgorodian+Shapiro+1974.
13. Morzy, Jozef. *Kryzys Demograficzny Na Litwie i Bialorusi w II Polowie XVII Wieku [the Demographic Crisis in Lithuania and Belarus in the Second Half of the 17th Century]*. Wydawnictwo Naukowe Uniwersytetu im. Adama Mickiewicza, 1965, https://search.worldcat.org/search?q=Demographic+Crisis+Lithuania+Belarus+Second+Half+17th+Century+Morzy+1965.
14. Seppel, Marten. “The Assistance of Manors to Peasants in Cases of Subsistence Crisis in Seventeenth-Century Estonia and Livonia.” *Acta Historica Tallinnensia*, vol. 8, 2004, pp. 3-19, https://doi.org/10.3176/hist.2004.1.01.
15. Malve, Martin. “First Glimpse of Infectious Diseases in Medieval and Early Modern Estonia Based on

Osteoarchaeological Material." *Estonian Journal of Archaeology*, vol. 27, no. 3S, 2023, pp. 165-84, https://doi.org/10.3176/arch.2023.3S.08.

16. Majewska, Gabriela. "The Polish-Lithuanian Commonwealth and Its Inhabitants During the Great Northern War (1700-1721)." *Studia Maritima*, vol. 31, 2018, pp. 125-46, https://doi.org/10.18276/sm.2018.31-07.
17. Bagaley, Dmitriy Ivanovich. *Ocherki Iz Istorii Kolonizatsii i Byta Stepnoy Okrainy Moskovskogo Gosudarstva [Essays on the History of Colonisation and Life on the Steppe Frontier of the Muscovite State]*. Universitetskaya Tipografiya, 1887, https://search.worldcat.org/search?q=Essays+History+Colonisation+Life+Steppe+Frontier+Muscovite+State+Bagaley+1887.
18. Stevens, Carol Belkin. *Soldiers on the Steppe: Army Reform and Social Change in Early Modern Russia*. Northern Illinois University Press, 1995, https://search.worldcat.org/search?q=Soldiers+Steppe+Army+Reform+Social+Change+Early+Modern+Stevens+1995.
19. Moon, David. *The Russian Peasantry 1600-1930: The World the Peasants Made*. Longman, 1999, https://doi.org/10.1093/ehr/116.465.229.
20. Pokrovsky, Nikolay Nikolaevich. *Tomsk 1648-1649 Gg.: Voevodskaya Vlast i Zemskie Miry [Tomsk in 1648-1649: Voivode Authority and Local Communities]*. Nauka, 1984, https://search.worldcat.org/search?q=Tomsk+1648+1649+Voivode+Authority+Local+Communities+Pokrovsky+1984.
21. Avdeev, Aleksandr Aleksandrovich. "Dorevizskiy Uchet Naseleniya v Rossii (XIII-XVII Veka): Istoriografiya i Istochniki [Pre-Revision Population Registration in Russia (13th-17th Centuries): Historiography and Sources]." *Istoricheskaya Demografiya*, vol. 1, 2025, pp. 12-38, https://doi.org/10.17323/demreview.v12i1.26573.
22. Ulyanova, Galina Nikolaevna. "Revizskie Skazki Kak Istochnik v Istoriografii 1970-Kh - 2010-Kh Gg. [Revision Tales as a Source in the Historiography of the 1970s-2010s]." *Istoricheskiy Zhurnal: Nauchnye Issledovaniya*, no. 4, 2016, pp. 412-25, https://cyberleninka.ru/article/n/revizskie-skazki-kak-istochnik-v-istoriografii-1970-h-2010-h-godov.
23. Shapiro, Aleksandr Lvovich. *Agrarnaya Istoriya Severo-Zapada Rossii: Vtoraya Polovina XV - Nachalo XVI v. [Agrarian History of North-Western Russia: Second Half of the 15th to Early 16th Century]*. Nauka, 1971, https://search.worldcat.org/search?q=Agrarian+History+North+Western+Russia+Second+Half+15th+Shapiro+1971.
24. Mironov, Boris Nikolaevich. *Novaya Istoricheskaya Demografiya Imperskoy Rossii [New Historical Demography of Imperial Russia]*. Dmitriy Bulanin, 2007, https://cyberleninka.ru/article/n/novaya-istoricheskaya-demografiya-imperskoy-rossii-analiticheskiy-obzor-sovremennoy-literatury-1.
25. Vahtre, Sulev. *Eestimaa Talurahvas Hingeloenduste Andmeil (1782-1858) [the Peasantry of Estland According to the Soul Revisions and Earlier Land Cadastres]*. Eesti Raamat, 1973, https://search.worldcat.org/search?q=Peasantry+Estland+According+Soul+Revisions+Earlier+Land+Cadastres+Vahtre+1973.
26. Lowmianski, Henryk. *Studia Nad Dziejami Wielkiego Ksiestwa Litewskiego [Studies on the History and Demography of the Grand Duchy of Lithuania]*. Wydawnictwo Naukowe UAM, 1983, https://search.worldcat.org/search?q=Studies+History+Demography+Grand+Duchy+Lithuania+Lowmianski+1983.
27. Krypiakevych, Ivan Petrovych. *Bohdan Khmelnytskyi [Bohdan Khmelnytsky and the Social-Demographic Transformation of 17th-Century Ukraine]*. Vydavnytstvo AN URSR, 1954, https://search.worldcat.org/search?q=Bohdan+Khmelnytsky+Social+Demographic+Transformation+17th+Century+Ukraine+Krypiakevych+1954.
28. Ozturk, Yucel. "An Eastern European Capital City: Kefe and the Demography of the Crimean Khanate and Ottoman Eyâlet (1475-1700)." *Karadeniz Arastirmalari*, vol. 16, no. 62, 2019, pp. 185-212, https://doi.org/10.22378/kio.2019.1.45-70.
29. Bernotas, Rivo. "Estonian Small Towns in the Middle Ages: Archaeology and the History of Urban

Defense." *Ajalooline Ajakiri / The Estonian Historical Journal*, no. 3, 2013, pp. 265-97, https://doi.org/10.12697/AA.2013.3.01.

30. Karjaharm, Toomas. *Ida Ja Laane Vahel: Eesti-Vene Suhted [Between East and West: Estonian-Russian Relations and Baltic Demography]*. Eesti Entsuklopeediakirjastus, 1998, https://search.worldcat.org/search?q=Between+East+West+Estonian+Russian+Relations+Baltic+Demography+Karjaharm+1998.
31. Plath, Ulrike, and Kaarel Vanamolder. "Clashing Weatherlands in 17th-Century Livonia: Climate, Subsistence and the Great Famine of 1695-1697." *Methis: Studia Humaniora Estonica*, vol. 24, no. 30, 2022, pp. 38-64, https://doi.org/10.7592/methis.v24i30.22101.
32. Rachuba, Andrzej. "Livonian and Courlander Nobility in Samogitia in the 17th and 18th Century." *Klio: Czasopismo Poswiecone Dziejom Polski i Powszechnym*, vol. 55, no. 3, 2020, pp. 85-110, https://doi.org/10.12775/klio.2020.039.
33. Petersone-Gordina, Elina, and Guntis Gerhards. "Dental Disease and Urban Mortality in a 17th-18th Century Community in Jelgava (Mitau), Duchy of Courland." *Papers on Anthropology*, vol. 20, 2011, pp. 311-26, https://scholar.google.com/scholar?q=Dental+Disease+Urban+Mortality+17th+18th+Century+Community+Petersone+2011.
34. Janicki, Arkadiusz. "Russian Expansion in the Baltic in the 18th Century: Demographic Losses of the Great Northern War and Plague." *Studia Historica Gedanensia*, vol. 13, 2022, pp. 101-23, https://doi.org/10.4467/23916001hg.22.006.17426.
35. Kiaupa, Zigmantas, et al. *The History of Lithuania Before 1795*. Lithuanian Institute of History, 2000, https://search.worldcat.org/search?q=History+Lithuania+Before+1795+Kiaupa+2000.
36. Vasiliauskas, Ernestas. "The Development of Towns of the Siauliai Crown Estate in Samogitia from the 16th to the 18th Century." *Archaeologia Baltica*, vol. 18, 2012, pp. 186-205, https://doi.org/10.15181/ab.v16i0.35.
37. Zvirblys, Atas. "Early Modern Urban Material Culture and Trade Connections in 17th-Century Vilnius." *Estonian Journal of Archaeology*, vol. 25, no. 2, 2021, pp. 93-112, https://doi.org/10.3176/arch.2021.2.01.
38. Katajala, Kimmo. "Liniya, Zona Ili Resheto? Kontseptualizatsiya Rossiysko-Shvedskoy Granitsy 1617 g. [Line, Zone or Sieve? Conceptualising the Russo-Swedish Border of Stolbovo 1617 in Kexholm and Karelia]." *Nordic and Baltic Studies Review*, no. 2, 2017, pp. 42-65, https://doi.org/10.15393/j103.art.2017.755.
39. Selin, Adrian Aleksandrovich. "Izgnannye Iz Livonii: Ob Ustroystve Pomeshchikov Starykh i Novykh v Novgorodskoy Zemle v Kontse XVI v. [Exiled from Livonia: The Resettlement of Service Landholders in the Novgorod Land in the Late 16th Century]." *Novgorodskiy Istoricheskiy Sbornik*, vol. 19, 2020, pp. 175-98, https://doi.org/10.21638/spbu19.2020.202.
40. Fishman, Olga Mikhaylovna. "The Ethnography and Demography of Karelians Outside Karelia: Tver, Tikhvin and Valdai Resettlement." *Nordic and Baltic Studies Review*, no. 5, 2020, pp. 218-42, https://doi.org/10.15393/j103.art.2020.1648.
41. Stepanova, Yuliya Vladimirovna. "Pervye Karelskie Vykhodtsy v Pogostakh Tverskoy Poloviny Bezhetskoy Pyatiny Po Dannym Pistsovykh Opisaniy Vtoroy Chetverti XVII v. [The First Karelian Migrants in the Tver Half of Bezhetskaya Pyatina According to Cadastres of the Second Quarter of the 17th Century]." *Ezhegodnik Finno-Ugorskikh Issledovaniy*, vol. 18, no. 1, 2024, pp. 102-16, https://doi.org/10.35634/2224-9443-2024-18-3-435-445.
42. Tikhomirov, Mikhail Nikolaevich. *Rossiya v XVI Stoletii [Russia in the 16th Century]*. Izdatelstvo Akademii Nauk SSSR, 1962, https://search.worldcat.org/search?q=Russia+16th+Century+Tikhomirov+1962.
43. Tsybakov, Denis Leonidovich. "Problemy Opredeleniya Vladeteley Territorii Verkhney Oki Na Rubezhe XV-XVI Vv. [Problems of Identifying Rulers and Principalities of the Upper Oka Territory at

the Turn of the 15th-16th Centuries]." *Istoriya: Fakty i Simvoly*, no. 3, 2022, pp. 44-56, https://doi.org/10.52070/2500-347X_2022_4_849_110.

44. Postnikov, Arseniy Borisovich. "Posadskie Lyudi Pskova v Istoriografii [the Posad Population of Pskov in Historiography]." *Pskov: Nauchno-Prakticheskiy, Istoriko-Kraevedcheskiy Zhurnal*, no. 54, 2021, pp. 28-41, https://cyberleninka.ru/article/n/posadskie-lyudi-pskova-v-istoriografii.
45. Bogdanov, Vladimir Olegovich. "Sukhoputnye Dorogi Uezda Rzhevy Volodimerovoy v XV-XVII Vv. [Overland Roads and Settlement Network of Rzheva Volodimerova Uezd in the 15th-17th Centuries]." *Istoricheskaya Geografiya*, vol. 7, 2025, pp. 104-36, https://doi.org/10.24888/2410-4205-2025-42-1-8-21.
46. Hellie, Richard. *Enserfment and Military Change in Muscovy*. University of Chicago Press, 1971, https://doi.org/10.7208/chicago/9780226326467.001.0001.
47. Davies, Brian L. *Warfare, State and Society on the Black Sea Steppe, 1500-1700*. Routledge, 2007, https://doi.org/10.4324/9780203961766.
48. Moiseev, Maksim Vladimirovich. "Stepnye Voyny Ot Ugry Do Nogayskogo Pogroma Kryma (1480-1523 Gg.) [Steppe Wars from the Stand on the Ugra to the Nogai Devastation of Crimea, 1480-1523]." *Zolotoordynskaya Tsivilizatsiya*, no. 8, 2015, pp. 252-74, https://cyberleninka.ru/article/n/stepnye-voyny-ot-ugry-do-nogayskogo-pogroma-kryma-1480-1522-gg.
49. Sharukho, Igor Nikolaevich. "Mestechki Belarusi: Iz Proshlogo v Budushchee [the Miasteczka of Belarus: Urban-Agrarian Settlement Evolution from the 15th to the 20th Century]." *Pskovskiy Regionologicheskiy Zhurnal*, no. 10, 2010, pp. 115-24, https://cyberleninka.ru/article/n/mestechki-belarusi-iz-proshlogo-v-buduschee.
50. Hellie, Richard. *The Economy and Material Culture of Russia, 1600-1725*. University of Chicago Press, 1999, https://doi.org/10.2307/2652607.
51. Zorin, Aleksandr Vasilevich. "Aleksandr Yuzef Lisovskiy: Geroy Smutnogo Vremeni [Aleksander Jozef Lisowski and the Raids of the Time of Troubles in Seversk and Central Russia]." *Istoriya Voenno-Politicheskikh Konfliktov*, vol. 4, 2012, pp. 1-78, https://cyberleninka.ru/article/n/aleksandr-yuzef-lisovskiy-geroy-smutnogo-vremeni.
52. Papkov, Andrey Igorevich. "Osvoenie Rossiyskim Gosudarstvom Yuzhnykh Okrain v XVI-XVII Vekakh [Colonisation of the Southern Borderlands by the Russian State in the 16th-17th Centuries]." *Nauchnye Vedomosti Belgorodskogo Gosudarstvennogo Universiteta: Istoriya, Politologiya*, vol. 25, no. 1, 2013, pp. 82-90, https://cyberleninka.ru/article/n/osvoenie-rossiyskim-gosudarstvom-yuzhnyh-okrain-v-xvi-veke.
53. Irtenina, Natalya Valerevna. "Evolyutsii Malorossiyskogo Kazachestva v XVII-XVIII Vv. [Social and Demographic Evolution of the Left-Bank Ukrainian Cossack Hetmanate in the 17th-18th Centuries]." *Voprosy Istorii*, no. 6, 2022, pp. 92-107, https://cyberleninka.ru/article/n/evolyutsii-malorossiyskogo-kazachestva-v-xviii-nachale-xx-v-iz-istorii-odnogo-roda.
54. Papkov, Andrey Igorevich. "Rasselenie Cherkas Na Yuzhnoy Okraine Rossii v Kontse XVI - Nachale XVIII Veka [the Settlement of Ukrainian Cherkasy on the Southern Frontier of Russia from the Late 16th to the Early 18th Century]." *Rusin*, no. 3, 2013, pp. 55-71, https://cyberleninka.ru/article/n/rasselenie-cherkas-na-yuzhnoy-okraine-rossii-v-kontse-xvi-nachale-xvii-v.
55. Krasnova, Irina Vladimirovna. "Istoki Formirovaniya Slobozhanshchiny Kak Chasti Geokulturnogo Prostranstva Rossii i Ukrainy [Origins of the Formation of Sloboda Ukraine as a Frontier Demographic Space]." *Via in Tempore: Istoriya, Politologiya*, vol. 47, no. 2, 2020, pp. 358-67, https://doi.org/10.34685/HI.2020.26.95.011.
56. Zhigalov, Vladimir Mikhaylovich. "Formirovanie i Periodizatsiya Voenno-Administrativnoy Sistemy Belgorodskogo Razryada v Kontse XVI - XVII Vv. [Formation and Periodisation of the Military-Administrative System of the Belgorod Razryad in the Late 16th-17th Centuries]." *Izvestiya*

Voronezhskogo Gosudarstvennogo Pedagogicheskogo Universiteta, no. 1, 2024, pp. 64-73, https://doi.org/10.52575/2687-0967-2024-51-1-102-115.

57. Kabuzan, Vladimir Maksimovich. *Izmeneniya v Razmeshchenii Naseleniya Rossii v XVIII - Pervoy Polovine XIX v. [Changes in the Distribution of the Population of Russia in the 18th and First Half of the 19th Century]*. Nauka, 1971, https://search.worldcat.org/search?q=Changes+Distribution+Population+Russia+18th+First+Half+19th+Kabuzan+1971.
58. Astafev, Dmitriy Aleksandrovich. "Osobennosti Izucheniya Genealogii Odnodvortsev Yuzhnoy Rossii XVII-XVIII Vv. [Specifics of Studying the Genealogy and Cadastral Registration of Odnodvortsy in Southern Russia in the 17th-18th Centuries]." *Istoricheskiy Vestnik*, vol. 39, 2022, pp. 118-35, https://doi.org/10.18287/2542-0445-2022-28-4-24-32.
59. Kurnikova, Oksana Mikhaylovna. "Krepost Taman v Strukture Osmanskoy Provintsii Kefe v XVI-XVIII Vekakh [Fortress Taman in the Structure of the Ottoman Province of Kefe in the 16th-18th Centuries]." *Materialy Po Arkheologii, Istorii i Etnografii Tavrii*, vol. 31, 2026, pp. 210-28, https://doi.org/10.31696/2072-5795-2026-2-5-11.
60. Tepkeev, Vladimir Toltaevich. "Kalmytsko-Dzhungarskie Otnosheniya i Kochevya Kalmytskogo Khanstva v Nizhnem Povolzhe v Kontse XVII Veka [Kalmyk-Dzungar Relations and the Nomadic Pastures of the Kalmyk Khanate on the Lower Volga in the Late 17th Century]." *Oriental Studies*, vol. 18, no. 1, 2025, pp. 52-66, https://doi.org/10.54770/20729286-2025-2-257.
61. Prudnikov, Yaroslav Vladimirovich. "Donskoy i Donetskiy Kazachiy Gorodok XVI-XVII Vv. [Don and Donets Cossack Fortified Settlements in the 16th-17th Centuries]." *Izvestiya Vuzov: Severo-Kavkazskiy Region*, no. 2, 2024, pp. 38-49, https://doi.org/10.15688/jvolsu4.2024.2.4.
62. Albogachiev, Magomed Mikhaylovich. "Akkinskie Migratsii i Rasselenie Vakhnakhov v XV-XVII Vv. [Vainakh and Akkin Migrations and Piedmont Resettlement in the 15th-17th Centuries]." *Vestnik Instituta Istorii, Arkheologii i Etnografii DNTs RAN*, vol. 21, no. 1, 2025, pp. 74-89, https://doi.org/10.7256/2454-0609.2025.2.70633.
63. Dadaev, Yusup Usmanovich. "Sistema Organov Vlasti i Upravleniya Feodalnykh Vladeniy i Soyuzov Selskikh Obshchestv Dagestana v XV-XVIII Vv. [System of Governance in Feudal Polities and Unions of Rural Jamaats of Dagestan in the 15th-18th Centuries]." *Vestnik Dagestanskogo Nauchnogo Tsentra*, no. 53, 2014, pp. 82-91, https://cyberleninka.ru/article/n/sistema-organov-vlasti-i-upravleniya-feodalnyh-vladeniy-dagestana.
64. Mustafina, Dina Abdulbarovna. "Materialy Opisaniy 1685-1687 Gg. Kak Istochnik Po Istorii Zemlevladeniya v Kazanskom Uezde [Materials of the 1685-1687 Surveys as a Source on Landholding in Kazan Uezd]." *Uchenye Zapiski Kazanskogo Universiteta*, vol. 150, no. 1, 2008, pp. 88-99, https://cyberleninka.ru/article/n/materialy-opisaniy-1685-1687-gg-kak-istochnik-po-istorii-zemlevladeniya-v-kazanskom-uezde.
65. Belyakov, Andrey Vasilevich. "Nevidimki Russkoy Armii XVI Veka [Service Tatars of Kasimov, Kadom and Temnikov in the 16th-Century Russian Army]." *Zolotoordynskoe Obozrenie*, no. 2, 2013, pp. 145-62, https://cyberleninka.ru/article/n/nevidimki-russkoy-armii-xvi-veka.
66. Belyakov, Andrey Vasilevich. "Temnikovskoe Knyazhestvo Po Istochnikam XVI-XVII Vv. [The Temnikov Principality According to 16th-17th Century Sources]." *Srednevekovye Tyurko-Tatarskie Gosudarstva*, no. 6, 2014, pp. 31-44, https://cyberleninka.ru/article/n/temnikovskoe-knyazhestvo-po-istochnikam-xvi-xvii-vv.
67. Rezun, Dmitriy Yakovlevich, and Ruslan Sergeevich Vasilievsky. *Letopis Sibirskikh Gorodov [Chronicle of Siberian Towns]*. Novosibirskoe Knizhnoe Izdatelstvo, 1989, https://search.worldcat.org/search?q=Chronicle+Siberian+Towns+Rezun+1989.
68. Matveev, Andrey Vasilevich. "Puti Soobshcheniya Sibirskikh Khanstv [Overland and Riverine

Communication Routes of the Siberian Khanates in the 15th-16th Centuries]." *Vestnik Omskogo Universiteta*, no. 3, 2011, pp. 181-90, https://cyberleninka.ru/article/n/puti-soobscheniya-sibirskih-hanstv.

69. Vizgalov, Georgiy Petrovich, and Oleg Viktorovich Kardash. "Ostyatskaya Usadba v Posade Goroda Berezova XVII-XVIII Vv. [Indigenous and Russian Urban Households in the Posad of Berezov in the 17th-18th Centuries]." *Arkheologiya, Etnografiya i Antropologiya Evrazii*, no. 2, 2011, pp. 94-105, https://cyberleninka.ru/article/n/ostyatskaya-usadba-v-posade-goroda-berezova-xviii-v.
70. Solodkin, Yankel Gutmanovich. "Kodskoe Knyazhestvo i Russkie Vlasti Sibiri v Kontse XVI - Pervoy Treti XVII v. [The Koda Khanty Principality and Russian Authorities in Siberia in the Late 16th and Early 17th Centuries]." *Vestnik Ugrovedeniya*, no. 3, 2015, pp. 112-21, https://cyberleninka.ru/article/n/kodskoe-knyazhestvo-i-russkie-vlasti-sibiri-v-kontse-xvi-nachale-xvii-vv.
71. Solodkin, Yankel Gutmanovich. "Pervye Novokreshcheny Severo-Zapadnoy Sibiri (Konets XVI - Pervaya Tret XVII v.) [the First Baptised Indigenous Inhabitants and Garrison Settlements of North-Western Siberia]." *Vestnik Tomskogo Gosudarstvennogo Universiteta: Istoriya*, no. 58, 2019, pp. 25-31, https://cyberleninka.ru/article/n/pervye-novokrescheny-severo-zapadnoy-sibiri-konets-xvi-pervaya-tret-xvii-v.
72. Tychinskikh, Zaytuna Aptrashitovna. "Istoriko-Demograficheskaya Kharakteristika Tatarskogo Naseleniya Tobolskogo i Tyumenskogo Uezdov v XVII-XVIII Vv. [Historical-Demographic Profile of the Tatar Population of Tobolsk and Tyumen Uezds in the 17th-18th Centuries]." *Zolotoordynskoe Obozrenie*, vol. 7, no. 2, 2019, pp. 342-61, https://doi.org/10.17223/15617793/447/22.
73. Kolesnikov, Aleksandr Dmitrievich. *Russkoe Naselenie Zapadnoy Sibiri v XVIII - Nachale XIX Vv. [The Russian Population of Western Siberia in the 18th and Early 19th Centuries]*. Zapadno-Sibirskoe Knizhnoe Izdatelstvo, 1973, https://search.worldcat.org/search?q=Russian+Population+Western+Siberia+18th+Early+19th+Centuries+Kolesnikov+1973.
74. Kabuzan, Vladimir Maksimovich. *Narody Rossii v XVIII Veke: Chislennost i Etnicheskiy Sostav [Peoples of Russia in the 18th Century: Population Size and Ethnic Composition]*. Nauka, 1990, https://search.worldcat.org/search?q=Peoples+Russia+18th+Century+Population+Size+Ethnic+Composition+Kabuzan+1990.
75. Turaev, Vadim Anatolevich. "Rossiysko-Kitayskoe Pogranichye v XVII Veke [the Russian-Chinese Borderland in the Amur Basin in the 17th Century: Albazin, Nerchinsk and Daur-Ducher Demography]." *Rossiya i ATR*, no. 2, 2020, pp. 31-48, https://doi.org/10.24411/2658-5960-2020-10022.
76. Turaev, Vadim Anatolevich. "Khozhdenie Vstrech Solntsu v Kontekste Problem Prisoedineniya Dalnego Vostoka k Rossiyskomu Gosudarstvu (XVII-XVIII Vv.) [Moving Towards the Sun: Russian Annexation and Indigenous Demography of the Far East in the 17th-18th Centuries]." *Rossiya i ATR*, no. 1, 2013, pp. 5-21, https://cyberleninka.ru/article/n/hozhdenie-vstrech-solntsu-v-kontekste-problem-prisoedineniya-dalnego-vostoka-k-rossiyskomu-gosudarstvu-hvii-hviii-vv.
77. Ushnitskiy, Vasiliy Vasilevich. "Yakutskie Volosti v 30-40-Kh Godakh XVII Veka [Yakut Volosts and Yasak Assessments in the 1630s-1640s]." *Severo-Vostochnyy Gumanitarnyy Vestnik*, no. 3, 2017, pp. 26-34, https://cyberleninka.ru/article/n/yakutskie-volosti-v-30-40-h-godah-xvii-veka.
78. Tolkacheva, Natalya Vladimirovna. "Russkie Lyudi i Korennye Narody Kamchatki: Pervye Kontakty (Konets XVII - Nachalo XVIII v.) [Russian Servicemen and Indigenous Itelmen and Koryak Peoples of Kamchatka at the Turn of the 17th-18th Centuries]." *Vestnik Kemerovskogo Gosudarstvennogo Universiteta*, no. 4, 2015, pp. 85-92, https://doi.org/10.17217/2079-0333-2015-32-100-109.